

The genome as a mosaic. Clustering individuals using allele frequency and haplotype-based methods applied to human and microbial datasets. Identifying admixture using patterns of LD – its sources and its timings

Oscar Lao

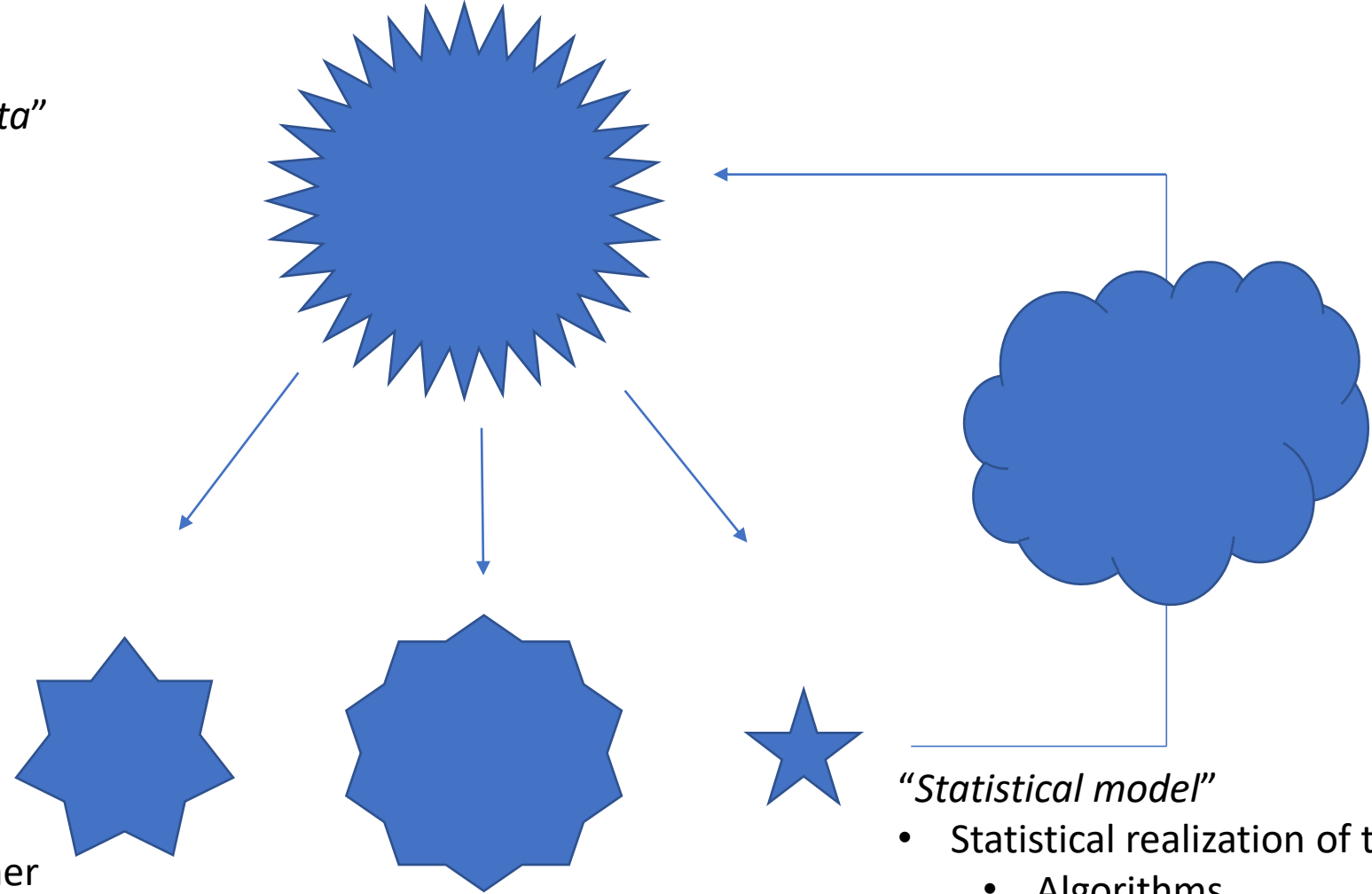
How to learn the “*model*”

“*Model that generates the data*”

- Conceptual
- Unseen
- Generalizes any dataset

“*Observed data*”

- Physical realization of the model (quantified features)
- Stochastic noise (it is not deterministic)
- Each dataset has particular features not shared with other datasets -> Each dataset is **UNIQUE**



“*Statistical model*”

- Statistical realization of the model
 - Algorithms

In **unsupervised** modelling

“Model that generates the data”

- Conceptual
- Unseen
- Generalizes any dataset

“Observed data”

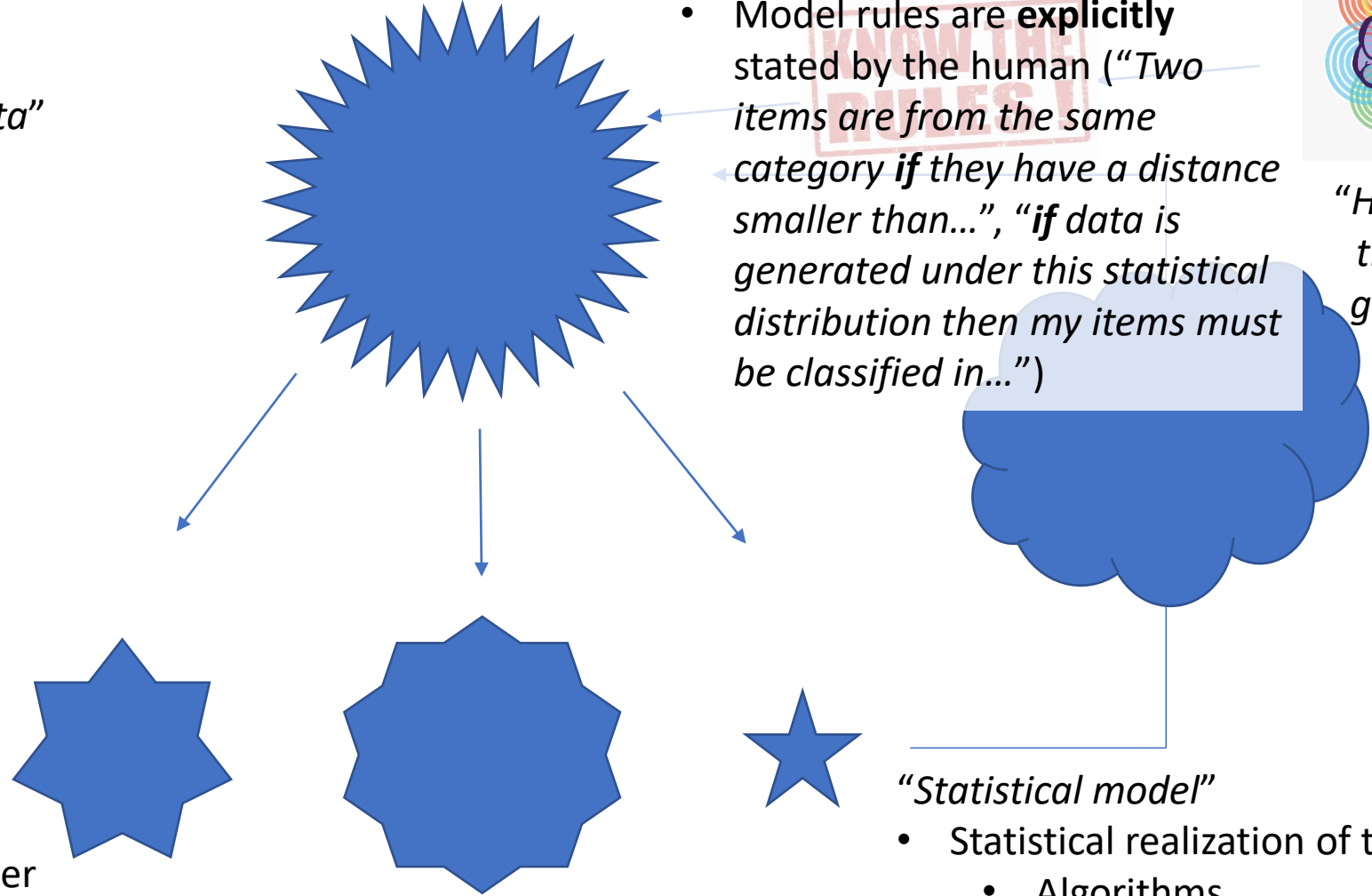
- Physical realization of the model (quantified features)
- Stochastic noise (it is not deterministic)
- Each dataset has particular features not shared with other datasets -> Each dataset is **UNIQUE**

- Model rules are **explicitly** stated by the human (“Two items are from the same category **if** they have a distance smaller than...”, “**if** data is generated under this statistical distribution then my items must be classified in...”)

“Human defines the rules that generated the concept”

“Statistical model”

- Statistical realization of the model
 - Algorithms



In **unsupervised** modelling

- Model rules are **explicitly** stated by the human (*“Two items are from the same category **if** they have a distance smaller than...”, “**if** data is generated under this statistical distribution then my items must be classified in...”*)

Where do they come from?

In **unsupervised** modelling

Unsupervised

Classification



- Requires a **explicit model** that explains the rules how the data is generated
- “*Explicit model*” -> **prior knowledge** has been used to define the model (Hume)
- There is an algorithm that implements that explicit model. We can do future predictions

Prior knowledge: “Things that look similar share a common origin”



Similar things can be clustered together

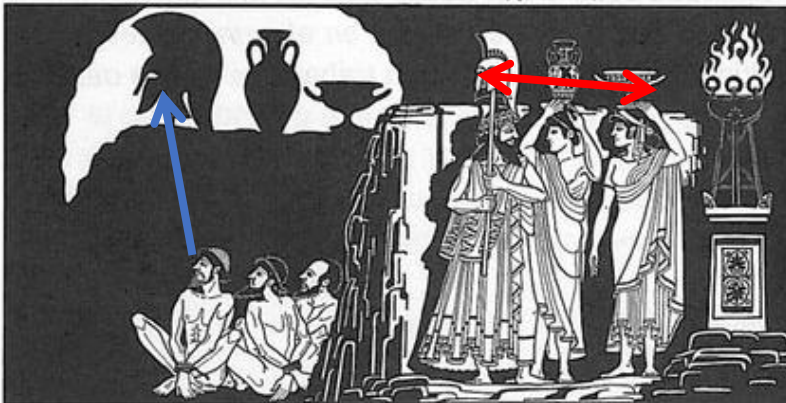
When to use unsupervised learning?

- Exploratory analyses
- Gain information about the inner structure of the data
- Dimensionality reduction
- Indirectly, to test hypotheses
- ...

Detecting population substructure is unsupervised modelling

How much can be detected and how much there is. The two sides of the same coin

Plato's cave myth



Detecting population substructure is unsupervised modelling

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Plato's cave myth



CHANGE THE ALGORITHM/HOW
DATA ARE DEFINED FOR DETECTING
POPULATION SUBSTRUCTURE

How data are defined?

Not all the features are good for pattern definition. Patterns are non-linear functions



Feature Selection vs feature Extraction

- **Feature extraction approach** -> features are projected into a new space with lower dimensionality.
 - Principle Component Analysis (PCA),
 - Linear Discriminant Analysis (LDA)
 - Singular Value Decomposition (SVD)
 - ...
- **Feature Selection approach** -> Identify a small subset of features that minimize redundancy and maximize relevance to the target (i.e., class label). FS ascertains features that are capable of discriminating samples that belong to different classes
 - Information Gain
 - Relief
 - Chi Squares
 - Fisher Score
 - Lasso
 - ...

Feature Selection vs feature Extraction

- Feature extraction approach -> features are projected into a new space with lower dimensionality.

Allows learning performance but interpretability of WHY the model Works is not so easy to identify

- Principle Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Singular Value Decomposition (SVD)
- ...

- Feature Selection approach -> Identify a small subset of features that minimize redundancy and maximize relevance to the target (i.e., class label). FS ascertains features that are capable of discriminating samples that belong to different classes

Allows better interpretability but pattern identification is constrained to the actual values of the ascertained features

- Information Gain
- Relief
- Chi Squares
- Fisher Score
- Lasso
- ...

Feature extraction

- Can we visualize the relationship between individuals given a SNV profile?

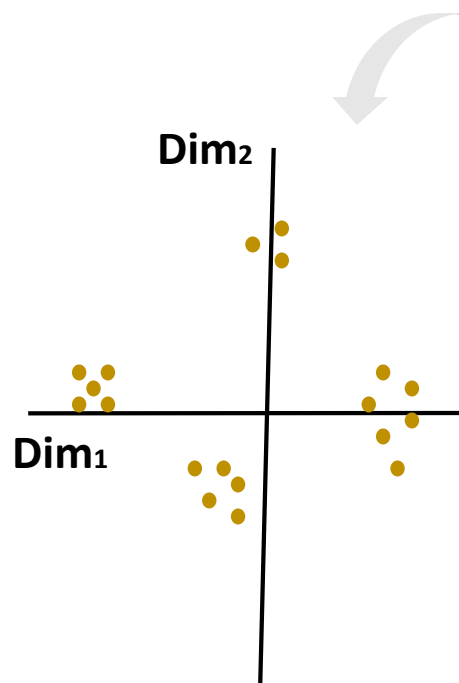
Feature extraction

- Can we visualize the relationship between individuals given a SNV profile?

Can we reduce the dimensionality of the genetic data so we can visualize the genetic relationships of individuals?
Can we identify any pattern in this reduced space?

Feature extraction

Assumptions of the model: “Features are (linear/non-linear correlated)”

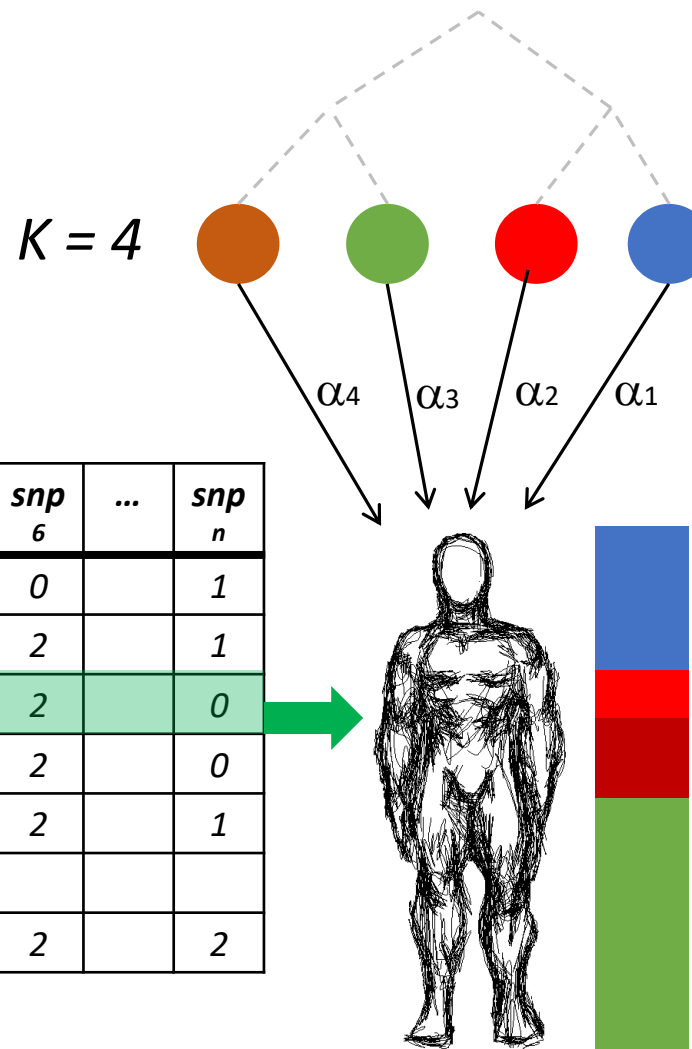


Algorithmic ancestry estimation

	snp ₁	snp ₂	snp ₃	snp ₄	snp ₅	snp ₆	...	snp _n
l ₁	0	1	2	1	0	0		1
l ₂	0	0	2	0	0	2		1
l ₃	1	1	2	0	0	2		0
l ₄	0	1	2	0	1	2		0
l ₅	1	1	2	0	1	2		1
...								
l _m	2	0	1	0	1	2		2

AA = 0
AB = 1
BB = 2

Assumptions of the model: “all ancestral populations are equally related”



Model-based ancestry estimation

Feature extraction

Assumptions of the model: “genetic relationships are a function of 2D space”



Algorithmic geographic estimation

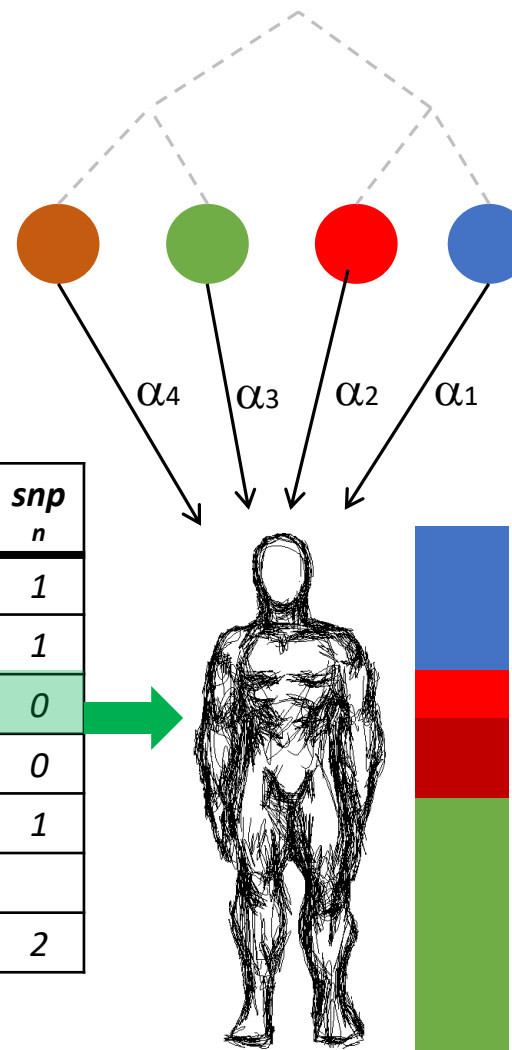
	<i>snp</i> ₁	<i>snp</i> ₂	<i>snp</i> ₃	<i>snp</i> ₄	<i>snp</i> ₅	<i>snp</i> ₆	...	<i>snp</i> _{<i>n</i>}
<i>l</i> ₁	0	1	2	1	0	0		1
<i>l</i> ₂	0	0	2	0	0	2		1
<i>l</i> ₃	1	1	2	0	0	2		0
<i>l</i> ₄	0	1	2	0	1	2		0
<i>l</i> ₅	1	1	2	0	1	2		1
...								
<i>l</i> _{<i>m</i>}	2	0	1	0	1	2		2

AA = 0

AB = 1

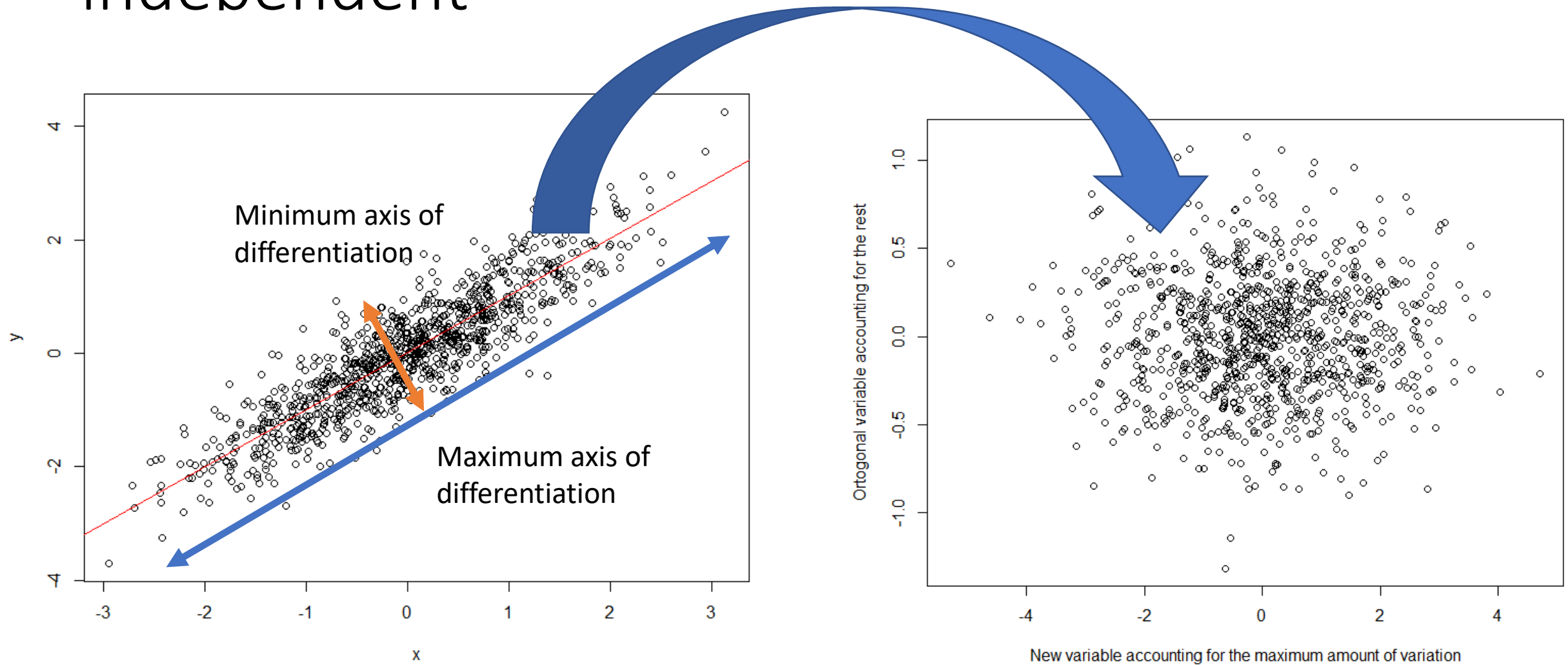
BB = 2

$K = 4$



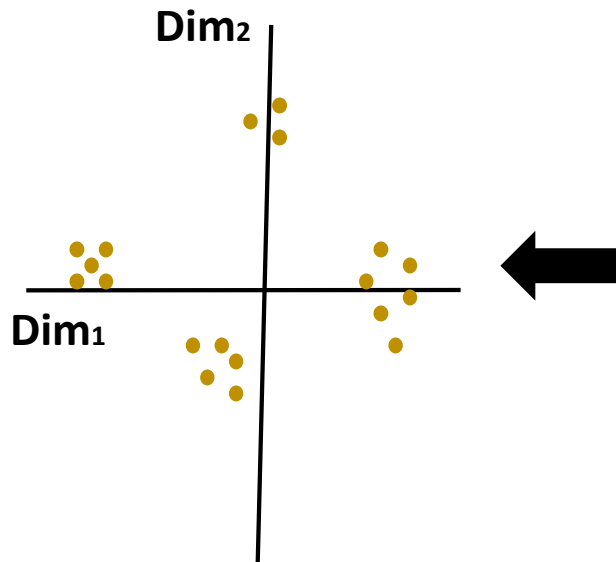
Model-based ancestry estimation

Feature Extraction: Not all variables are independent



Feature extraction: Change how data are defined

Patterns become more obvious



Feature extraction algorithm

Modify how data are defined. For example, data are defined NOT in terms of SNPs, but in terms of identical fragments (incorporate LD)

Identical By Descent	l_1	l_2	l_3	...	l_m
l_1					
l_2					
l_3					
...					
l_m					

LD

	<i>snp</i> ₁	<i>snp</i> ₂	<i>snp</i> ₃	<i>snp</i> ₄	<i>snp</i> ₅	<i>snp</i> ₆	...	<i>snp</i> _n
l_1	0	1	2	1	0	0		1
l_2	0	0	2	0	0	2		1
l_3	1	1	2	0	0	2		0
l_4	0	1	2	0	1	2		0
l_5	1	1	2	0	1	2		1
...								
l_m	2	0	1	0	1	2		2

$AA = 0$

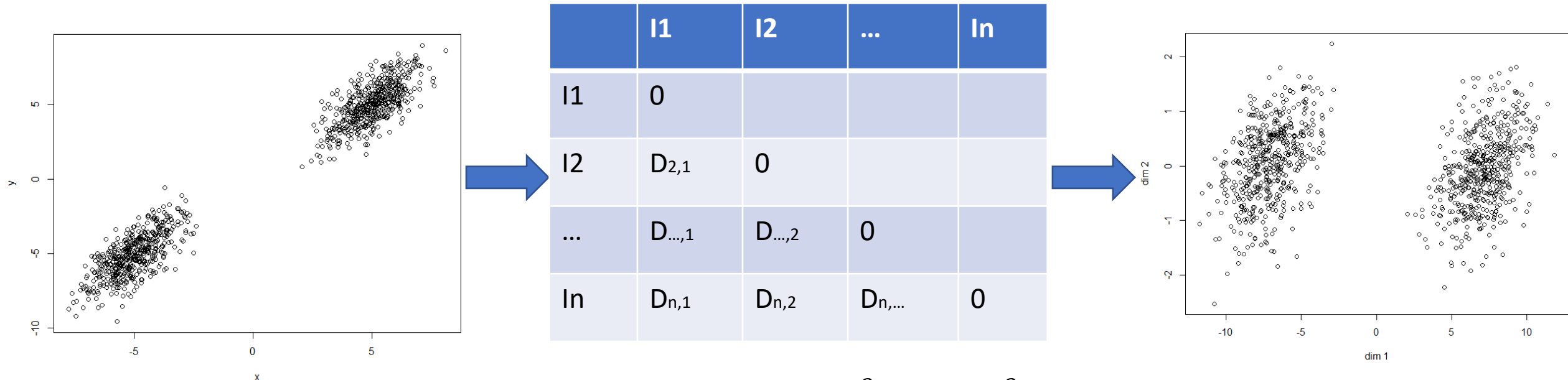
$AB = 1$

$BB = 2$

A coordinate analysis approach (classical multidimensional scaling)

$$D_{i,j} = \|x\|_2 = w_{dim1}(dim1_i - dim1_j)^2 + w_{dim2}(dim2_i - dim2_j)^2$$

$$w_{dim1} > w_{dim2}$$



$$d_{i,j} = \|x\|_2 = (x_i - x_j)^2 + (y_i - y_j)^2$$

$$d_{ij}(\mathbf{X}) = \left[\sum_{a=1}^m (x_{ia} - x_{ja})^2 \right]^{1/2} . \quad d_{rs}^2 = (\mathbf{x}_r - \mathbf{x}_s)^T (\mathbf{x}_r - \mathbf{x}_s).$$

The algorithm Coordinate Analysis (*classical multidimensional scaling*)

1. Compute the matrix of squared dissimilarities $\Delta^{(2)}$.
2. Apply double centering to this matrix:

$$\mathbf{B}_\Delta = -\frac{1}{2}\mathbf{J}\Delta^{(2)}\mathbf{J}. \quad (12.3)$$

3. Compute the eigendecomposition of $\mathbf{B}_\Delta = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}'$.
4. Let the matrix of the first m eigenvalues *greater than zero* be $\mathbf{\Lambda}_+$ and $\mathbf{Q}_+$ the first m columns of $\mathbf{Q}$. Then, the coordinate matrix of classical scaling is given by $\mathbf{X} = \mathbf{Q}_+\mathbf{\Lambda}_+^{1/2}$.

Interpretation of results

$$B_{\Delta} = Q\Lambda Q'$$

$$X = Q_+ \Lambda_+^{1/2}$$

Eigenvalues
(n-1)

Eigenvector
(n-1)

$$d_{rs}^2 = \sum_{i=1}^{n-1} \lambda_i (x_{ri} - x_{si})^2,$$

Positive

More weight

Less weight

Eigenvalue

$\lambda_1 = 11.47$

Eigenvector

(-1.16, -0.19, -0.07, 0.56, 1.01, -0.49, -0.71, -0.82, -0.42, -0.15, 0.26, -0.30, 0.40, -1.13, 0.02, -0.88, 0.45, 0.00, 0.11, -0.53, 0.79, -0.32, 0.37, -0.08, 0.09, 1.00, -0.41, 0.09, 0.47, 0.00, -0.01, -0.08, 0.60, 0.05, 0.60, 0.45, -0.23, -0.07, -0.24, 0.98)

$\lambda_2 = 4.98$

(0.11, -0.42, 0.21, -0.79, -0.14, -0.70, -0.26, -0.32, -0.03, -0.14, 0.00, 0.24, 0.14, 0.27, -0.64, 0.47, -0.51, -0.07, 0.36, -0.36, 0.31, 0.05, 0.28, -0.04, 0.38, -0.40, -0.33, 0.83, -0.19, -0.12, -0.01, -0.03, 0.26, 0.20, 0.22, 0.55, 0.16, 0.37, 0.40, 0.07)

$\lambda_3 = 4.56$

(-0.12, 0.15, -0.61, -0.10, -0.31, -0.07, -0.21, 0.33, -0.68, -0.01, 0.36, 0.56, -0.26, 0.07, -0.30, -0.16, -0.08, -0.02, -0.18, -0.30, -0.50, -0.69, -0.07, 0.06, 0.65, 0.34, 0.36, -0.25, 0.64, 0.49, 0.18, 0.30, -0.09, -0.02, 0.26, -0.20, 0.27, 0.45, -0.05, -0.19)

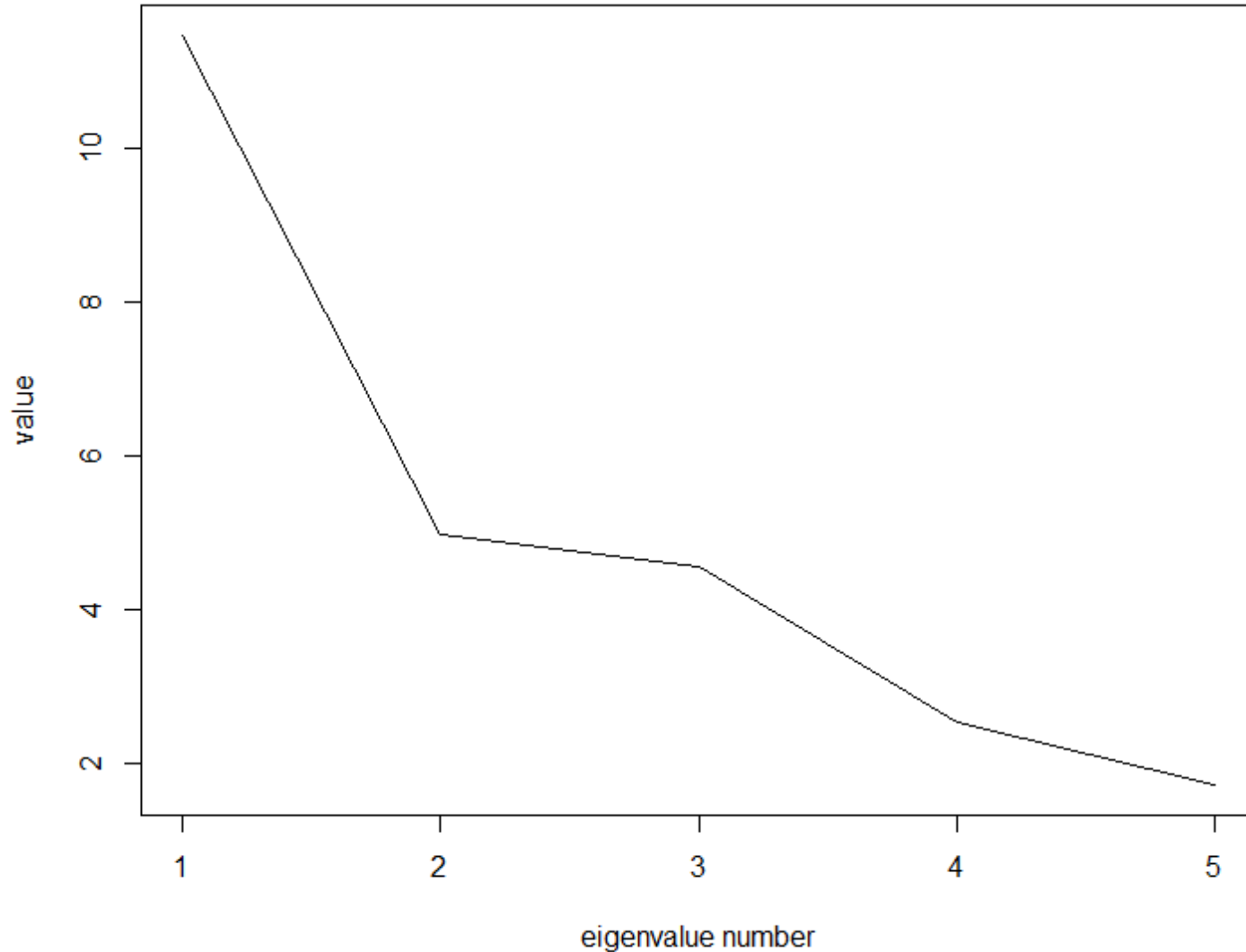
$\lambda_4 = 2.55$

(0.16, 0.04, -0.12, -0.12, 0.24, 0.15, 0.04, 0.20, 0.25, -0.16, -0.33, 0.39, 0.48, -0.20, -0.36, -0.07, 0.22, 0.53, -0.18, 0.02, 0.29, -0.55, 0.35, -0.15, -0.32, -0.19, 0.14, 0.10, 0.09, -0.27, 0.24, -0.05, 0.12, -0.09, 0.02, -0.15, -0.24, 0.17, -0.29, -0.44)

$\lambda_5 = 1.73$

(-0.03, -0.09, 0.23, 0.13, 0.07, -0.29, -0.11, 0.43, -0.08, -0.16, -0.04, -0.32, -0.18, 0.19, -0.37, -0.26, 0.32, 0.12, 0.17, 0.24, -0.20, -0.14, 0.11, 0.42, 0.15, -0.20, 0.05, 0.16, 0.06, 0.04, -0.25, -0.22, 0.40, 0.16, -0.25, -0.10, 0.09, -0.13, -0.10, 0.01)

Interpretation of results



Eigenvalue

$\lambda_1 = 11.47$

Eigenvector

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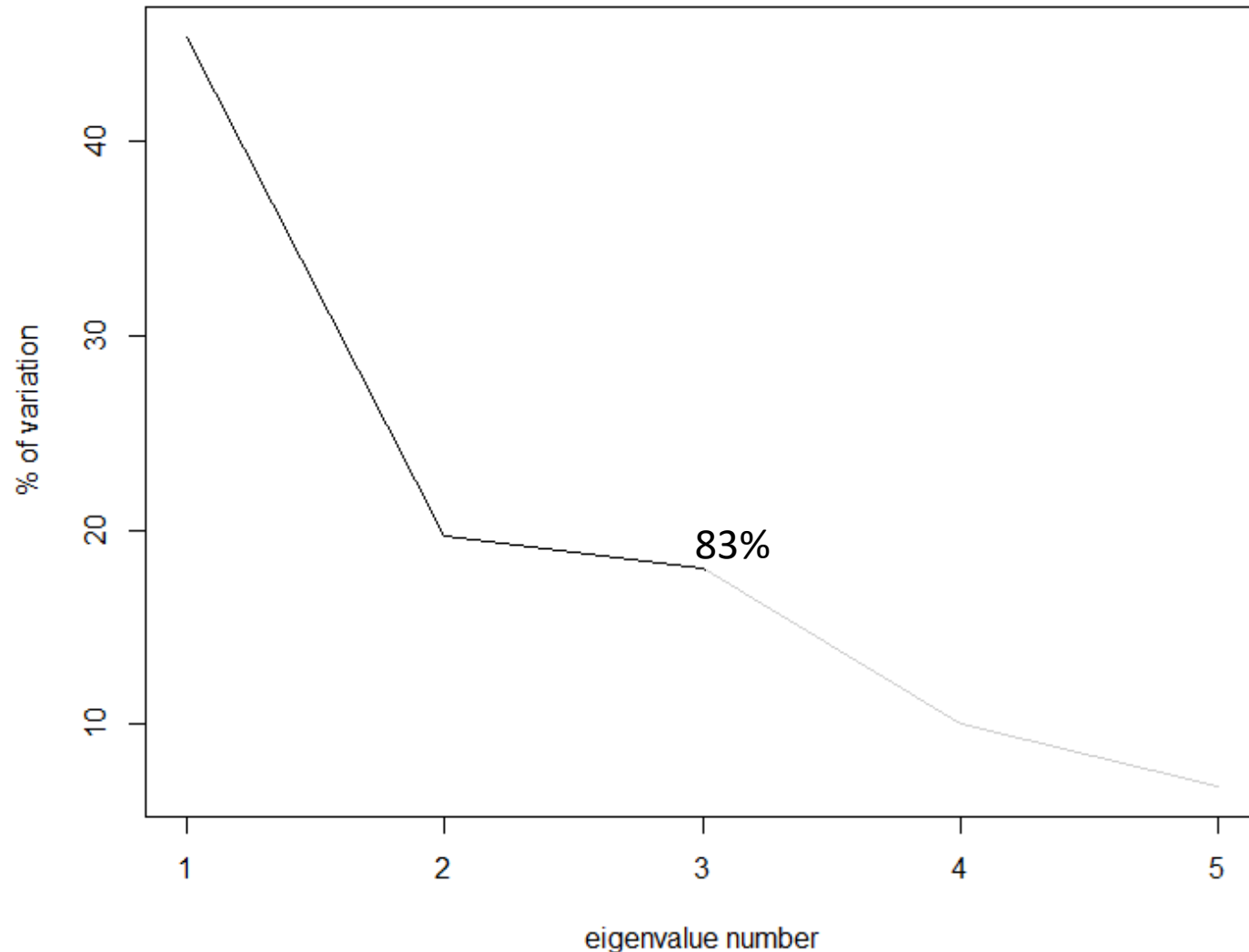
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Interpretation of results



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Many distances can be used!

Table 1.1 *Dissimilarity measures for quantitative data*

Euclidean distance	$\delta_{rs} = \left\{ \sum_i (x_{ri} - x_{si})^2 \right\}^{\frac{1}{2}}$
Weighted Euclidean	$\delta_{rs} = \left\{ \sum_i w_i (x_{ri} - x_{si})^2 \right\}^{\frac{1}{2}}$
Mahalanobis distance	$\delta_{rs} = \{(\mathbf{x}_r - \mathbf{x}_s)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_r - \mathbf{x}_s)\}^{\frac{1}{2}}$
City block metric	$\delta_{rs} = \sum_i x_{ri} - x_{si} $
Minkowski metric	$\delta_{rs} = \left\{ \sum_i w_i x_{ri} - x_{si} ^\lambda \right\}^{\frac{1}{\lambda}} \quad \lambda \geq 1$
Canberra metric	$\delta_{rs} = \sum_i x_{ri} - x_{si} / (x_{ri} + x_{si})$
Divergence	$\delta_{rs} = \frac{1}{p} \sum_i (x_{ri} - x_{si})^2 / (x_{ri} + x_{si})^2$
Bray-Curtis	$\delta_{rs} = \frac{1}{p} \frac{\sum_i x_{ri} - x_{si} }{\sum_i (x_{ri} + x_{si})}$
Soergel	$\delta_{rs} = \frac{\sum_i x_{ri} - x_{si} }{\sum_i \max(x_{ri}, x_{si})}$
Bhattacharyya distance	$\delta_{rs} = \left\{ \sum_i (x_{ri}^{\frac{1}{2}} - x_{si}^{\frac{1}{2}})^2 \right\}^{\frac{1}{2}}$
Wave-Hedges	$\delta_{rs} = \frac{1}{p} \sum_i \left(1 - \frac{\min(x_{ri}, x_{si})}{\max(x_{ri}, x_{si})} \right)$
Angular separation	$\delta_{rs} = 1 - \frac{\sum_i x_{ri} x_{si}}{[\sum_i x_{ri}^2 \sum_i x_{si}^2]^{\frac{1}{2}}}$
Correlation	$\delta_{rs} = 1 - \frac{\sum_i (x_{ri} - \bar{x}_r)(x_{si} - \bar{x}_s)}{\left\{ \sum_i (x_{ri} - \bar{x}_r)^2 \sum_i (x_{si} - \bar{x}_s)^2 \right\}^{\frac{1}{2}}}$

Error between the real distance matrix and the distance matrix computed with the new coordinates

$$e_{ij}^2 = (d_{ij} - \delta_{ij})^2$$

$$\sigma_r(\mathbf{X}) = \sum_{i=1}^n \sum_{j=i+1}^n (d_{ij} - \delta_{ij})^2 = \sigma_r(\mathbf{X}) = \sum_{i < j} (d_{ij} - \delta_{ij})^2 \quad \text{for all available } \delta_{ij}$$

Raw stress = $\sigma_r(\mathbf{X}) = \sum_{i < j} w_{ij} (d_{ij}(\mathbf{X}) - \delta_{ij})^2$.

{

$w_{ij} = 0$ if δ_{ij} is missing
 w_{ij} with value 1 if δ_{ij}

$$\begin{aligned} \sigma_r(\mathbf{X}) &= \sum_{i < j} w_{ij} (\delta_{ij} - d_{ij}(\mathbf{X}))^2 \\ &= \sum_{i < j} w_{ij} \delta_{ij}^2 + \sum_{i < j} w_{ij} d_{ij}^2(\mathbf{X}) - 2 \sum_{i < j} w_{ij} \delta_{ij} d_{ij}(\mathbf{X}) \\ &= \eta_\delta^2 + \eta^2(\mathbf{X}) - 2\rho(\mathbf{X}), \end{aligned}$$

STRUCTURE/ADMIXTURE principles

Ancestral pops

(Unseen) frequency of allele A
In one SNP

(Unseen) Genotype frequencies

Observed genotype in individual i

1



p_1

AA (0)

p_1^2

AB (1)

$2(1 - p_1)p_1$

BB (2)

$(1 - p_1)^2$

2



p_2

AA (0)

p_2^2

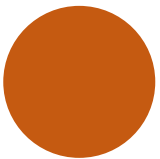
AB (1)

$2(1 - p_2)p_2$

BB (2)

$(1 - p_2)^2$

3



p_3

AA (0)

p_3^2

AB (1)

$2(1 - p_3)p_1$

BB (2)

$(1 - p_3)^2$

0



STRUCTURE/ADMIXTURE principles

Ancestral pops (Unseen) frequency of allele A
In one SNP (Unseen) Genotype frequencies Observed genotype in individual i

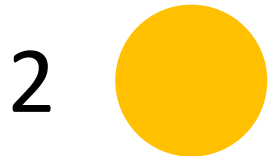


p_1

AA (0) p_1^2
AB (1) $2(1 - p_1)p_1$
BB (2) $(1 - p_1)^2$

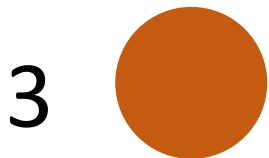
0

$$P(\text{pop} = 1 | G = 0) = \frac{P(\text{pop} = 1)p_1^2}{P(\text{pop} = 1)p_1^2 + P(\text{pop} = 2)p_2^2 + P(\text{pop} = 3)p_3^2}$$



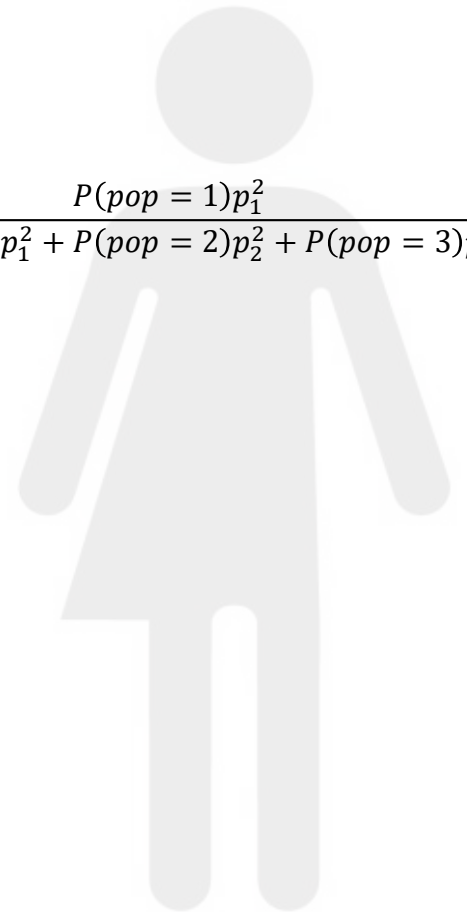
p_2

AA (0) p_2^2
AB (1) $2(1 - p_2)p_2$
BB (2) $(1 - p_2)^2$



p_3

AA (0) p_3^2
AB (1) $2(1 - p_3)p_1$
BB (2) $(1 - p_3)^2$



STRUCTURE/ADMIXTURE principles

If we have many SNVs, the probability of the vector of genotypes from a genome given a population is then computed as:

$$P(\text{Genome} | \text{pop} = i) = \prod_s^n P(G_s | \text{pop} = i)$$

Problem: what is the frequency of each SNP s in each population i ?

Solution: given k individuals, estimate the probabilities of each SNV at each ancestral population so the likelihoods are maximized. There are different methods, such as the Expectation-Maximization algorithm. Nowadays the algorithms that follow this framework differ on the optimization method that is applied!

ADMIXTURE is a particular case of MIXTURE OF PROBABILITY DISTRIBUTIONS!

Imagine a single dimension (y)

$$g_{\theta}(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(z-\mu)^2/\sigma^2}$$

$$Y_1 \sim N(\mu_1, \sigma_1^2),$$

$$Y_2 \sim N(\mu_2, \sigma_2^2),$$

$$Y = (1 - \Delta) \cdot Y_1 + \Delta \cdot Y_2, \quad \Delta \in \{0, 1\} \text{ with } \Pr(\Delta = 1) = \pi.$$

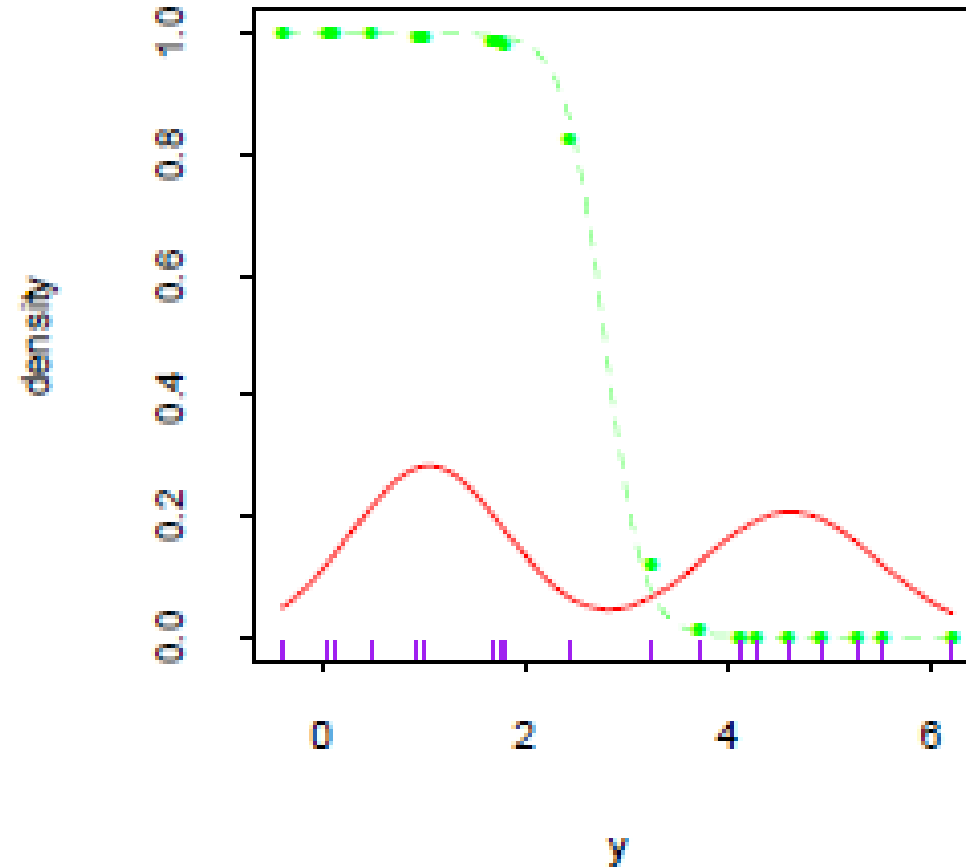
$$g_Y(y) = (1 - \pi)\phi_{\theta_1}(y) + \pi\phi_{\theta_2}(y).$$

$$\theta = (\pi, \theta_1, \theta_2) = (\pi, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2).$$

$$L(\theta; \mathbf{Z}) = \prod_{i=1}^N g_{\theta}(z_i),$$

$$\begin{aligned} \ell(\theta; \mathbf{Z}) &= \sum_{i=1}^N \ell(\theta; z_i) \\ &= \sum_{i=1}^N \log g_{\theta}(z_i), \end{aligned}$$

$$f(x) = \sum_{m=1}^M \alpha_m \phi(x; \mu_m, \Sigma_m)$$



Gaussian mixture and EM algorithm

π = probability of being of the group 2

$$\ell(\theta; \mathbf{Z}) = \sum_{i=1}^N \log[(1 - \pi)\phi_{\theta_1}(y_i) + \pi\phi_{\theta_2}(y_i)]. \quad \Delta \in \{0, 1\} \text{ with } \Pr(\Delta = 1) = \pi.$$

Direct maximization of $\ell(\theta; \mathbf{Z})$ is quite difficult numerically, because of the sum of terms inside the logarithm. There is, however, a simpler approach. We consider unobserved latent variables Δ_i taking values 0 or 1 as in (8.36): if $\Delta_i = 1$ then Y_i comes from model 2, otherwise it comes from model 1. Suppose we knew the values of the Δ_i 's. Then the log-likelihood would be

$$\begin{aligned} \ell_0(\theta; \mathbf{Z}, \Delta) &= \sum_{i=1}^N [(1 - \Delta_i) \log \phi_{\theta_1}(y_i) + \Delta_i \log \phi_{\theta_2}(y_i)] \\ &\quad + \sum_{i=1}^N [(1 - \Delta_i) \log(1 - \pi) + \Delta_i \log \pi], \end{aligned} \quad (8.40)$$

Since the values of the Δ_i 's are actually unknown, we proceed in an iterative fashion, substituting for each Δ_i in (8.40) its expected value

$$\gamma_i(\theta) = E(\Delta_i | \theta, \mathbf{Z}) = \Pr(\Delta_i = 1 | \theta, \mathbf{Z}), \quad (8.41)$$

Algorithm 8.1 *EM Algorithm for Two-component Gaussian Mixture.*

1. Take initial guesses for the parameters $\hat{\mu}_1, \hat{\sigma}_1^2, \hat{\mu}_2, \hat{\sigma}_2^2, \hat{\pi}$ (see text).
2. *Expectation Step*: compute the responsibilities

$$\hat{\gamma}_i = \frac{\hat{\pi} \phi_{\hat{\theta}_2}(y_i)}{(1 - \hat{\pi}) \phi_{\hat{\theta}_1}(y_i) + \hat{\pi} \phi_{\hat{\theta}_2}(y_i)}, \quad i = 1, 2, \dots, N. \quad (8.42)$$

3. *Maximization Step*: compute the weighted means and variances:

$$\begin{aligned} \hat{\mu}_1 &= \frac{\sum_{i=1}^N (1 - \hat{\gamma}_i) y_i}{\sum_{i=1}^N (1 - \hat{\gamma}_i)}, & \hat{\sigma}_1^2 &= \frac{\sum_{i=1}^N (1 - \hat{\gamma}_i) (y_i - \hat{\mu}_1)^2}{\sum_{i=1}^N (1 - \hat{\gamma}_i)}, \\ \hat{\mu}_2 &= \frac{\sum_{i=1}^N \hat{\gamma}_i y_i}{\sum_{i=1}^N \hat{\gamma}_i}, & \hat{\sigma}_2^2 &= \frac{\sum_{i=1}^N \hat{\gamma}_i (y_i - \hat{\mu}_2)^2}{\sum_{i=1}^N \hat{\gamma}_i}, \end{aligned}$$

and the mixing probability $\hat{\pi} = \sum_{i=1}^N \hat{\gamma}_i / N$.

4. Iterate steps 2 and 3 until convergence.
-

Gaussian mixture and EM algorithm

$$\hat{\gamma}_i = \frac{\hat{\pi} \phi_{\theta_2}(y_i)}{(1 - \hat{\pi}) \phi_{\theta_1}(y_i) + \hat{\pi} \phi_{\theta_2}(y_i)}, \quad i = 1, 2, \dots, N.$$

Probability that point i belongs to group 2 given the data

Prior probability of belonging to group 2

Likelihood of the data i in group 2

$$\hat{\mu}_1 = \frac{\sum_{i=1}^N (1 - \hat{\gamma}_i) y_i}{\sum_{i=1}^N (1 - \hat{\gamma}_i)}, \quad \longrightarrow \quad \text{Weighted mean for group 1 (1-probability of assigning to group 2)}$$

$$\hat{\mu}_2 = \frac{\sum_{i=1}^N \hat{\gamma}_i y_i}{\sum_{i=1}^N \hat{\gamma}_i}, \quad \longrightarrow \quad \text{Weighted mean for group 2}$$

$$\hat{\sigma}_1^2 = \frac{\sum_{i=1}^N (1 - \hat{\gamma}_i) (y_i - \hat{\mu}_1)^2}{\sum_{i=1}^N (1 - \hat{\gamma}_i)}, \quad \longrightarrow \quad \text{Weighted variance for group 1 (1-probability of assigning to group 2)}$$

$$\hat{\sigma}_2^2 = \frac{\sum_{i=1}^N \hat{\gamma}_i (y_i - \hat{\mu}_2)^2}{\sum_{i=1}^N \hat{\gamma}_i}, \quad \longrightarrow \quad \text{Weighted variance for group 2}$$

sNMF

For individual i and genotype G at SNPs, it can be considered a weighted average of the genotypes of the N ancestral populations

Total number of ancestries $\rightarrow$ N
 Genotype of individual i at SNP s $\rightarrow$ $G_s^i = \sum_{k=1}^N q^k G_s^k$
 Frequency of genotype s in population k $\rightarrow$ G_s^k
 Weight of ancestry k $\rightarrow$ q^k

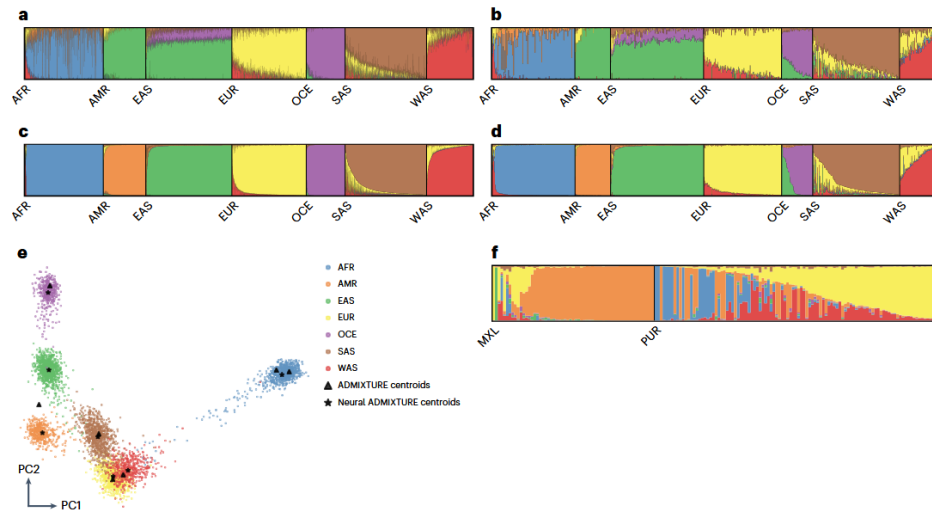
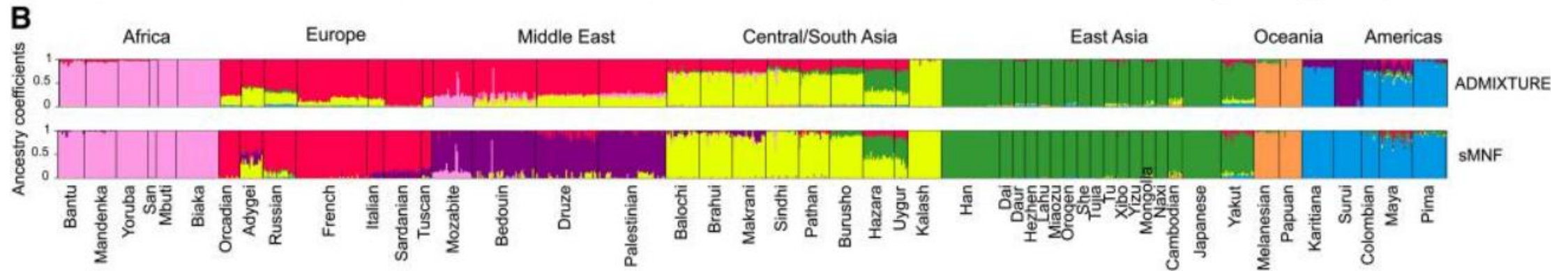
$$\begin{matrix} W \\ \begin{bmatrix} & & \\ & & \\ & & \\ & & \end{bmatrix} \end{matrix} \times \begin{matrix} H \\ \begin{bmatrix} & & & & & \\ & & & & & \end{bmatrix} \end{matrix} \approx \begin{matrix} V \\ \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{matrix}$$

To minimize

$$\|V - WH\|_F, \text{ subject to } W \geq 0, H \geq 0.,$$

Guess what: different algorithms produce different results...

Genetics. 2014 Feb 4;196(4):973–983

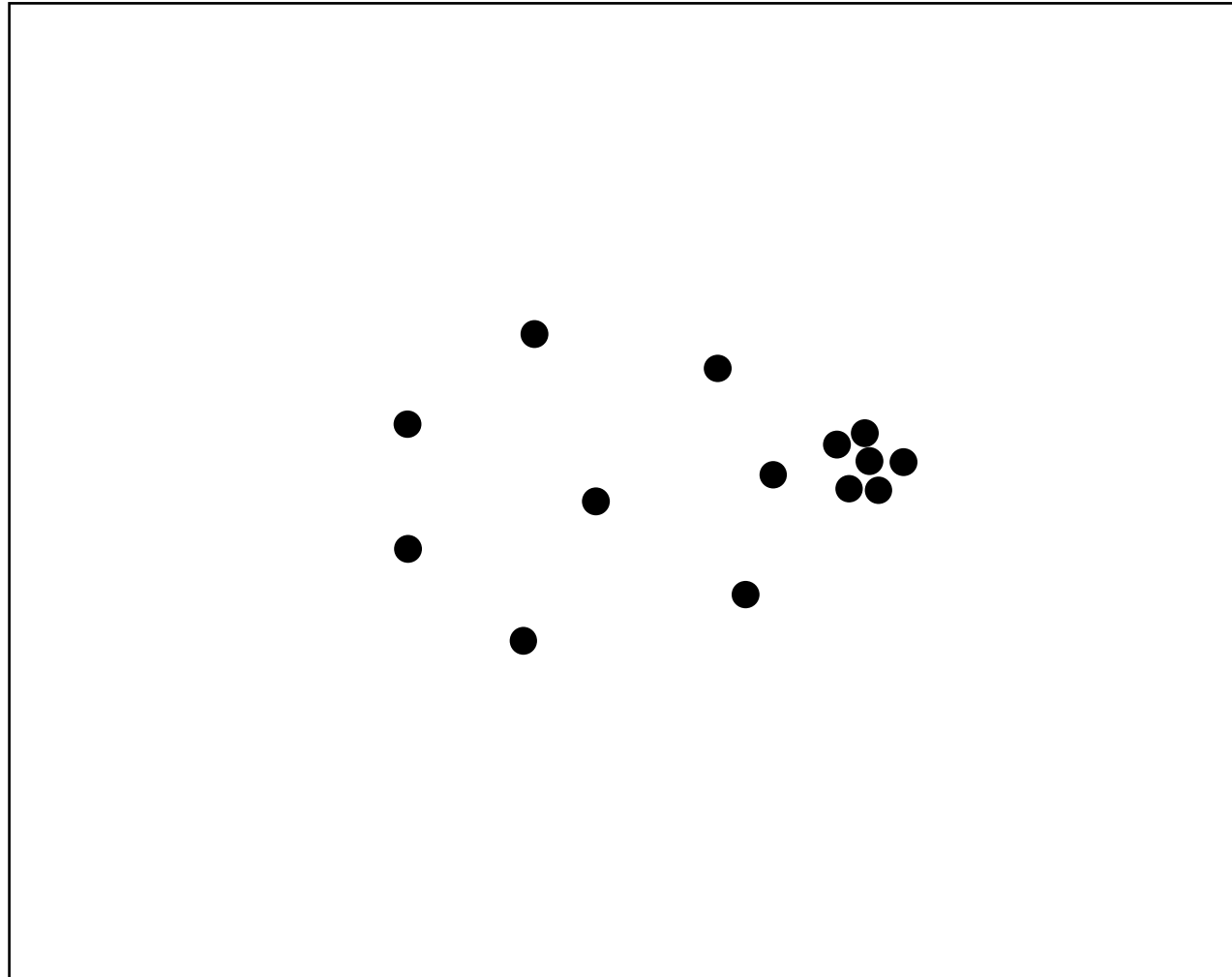


ADMIXTURE vs Neural ADMIXTURE

Global ancestry algorithms

Tool / Algorithm	Key Features	Input	Output	R Package / Software
STRUCTURE	Bayesian clustering, Q-matrix	Genotype matrix	Admixture proportions	structure
ADMIXTURE	Fast ML-based estimation	PLINK files	Admixture proportions	ADMIXTURE
fastSTRUCTURE	Scalable Bayesian via variational inference	Genotype matrix	Admixture proportions	fastSTRUCTURE (GitHub)
sNMF	Sparse non-negative matrix factorization	LFMM matrix	Ancestry proportions	LEA (CRAN)
PCA / MDS	Dimensionality reduction for structure visualization	SNP data	PCs per sample	SNPRelate or adeget
fineSTRUCTURE + ChromoPainter	Haplotype-based clustering via coancestry matrix	Phased haplotypes	Coancestry matrix & clusters	fineSTRUCTURE
FST-based methods	Pairwise FST to quantify divergence	Allele frequencies	FST matrix	hierfstat (CRAN)
t-SNE / UMAP	Non-linear embedding of PCA results	PCA scores	Cluster visualization	Rtsne , umap
Neural Admixture	Deep Learning modelling	Genotype matrix	Admixture proportions	NeuralAdmixture (GitHub)

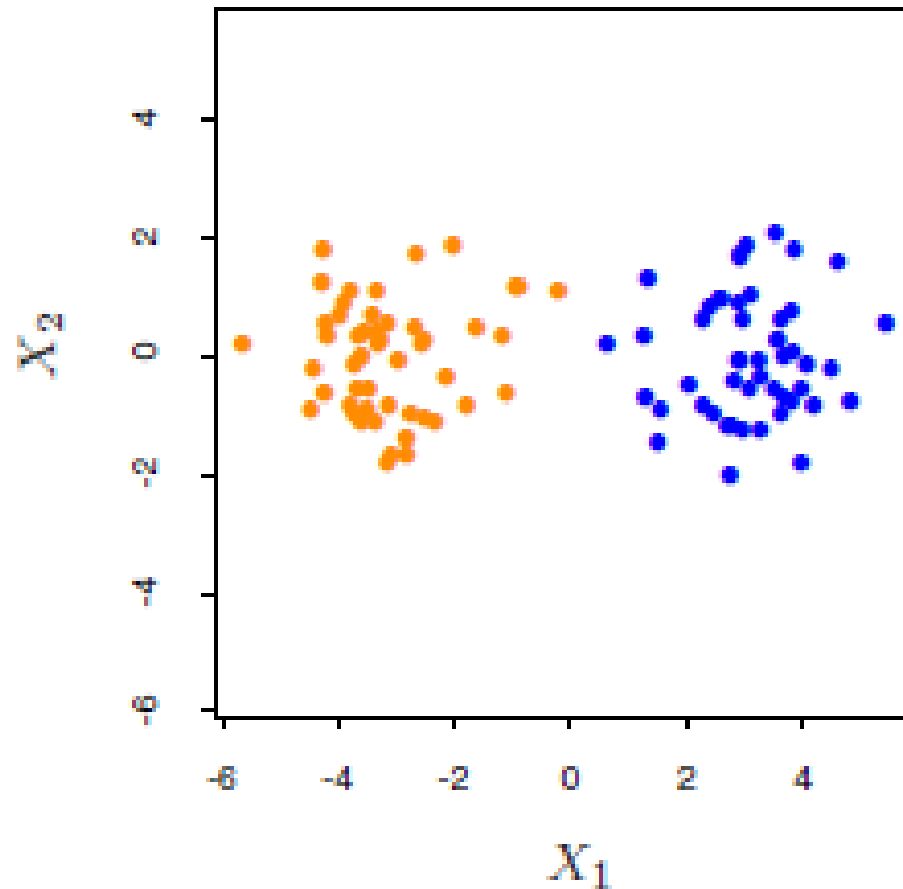
How many ancestral populations?



How many ancestral populations?

- Identifying the OPTIMAL number of clusters depends on:
- What do you understand by OPTIMAL?
- How do you measure such “optimality”?
- How do you explore the space of solutions to reach the optimal point (K)?

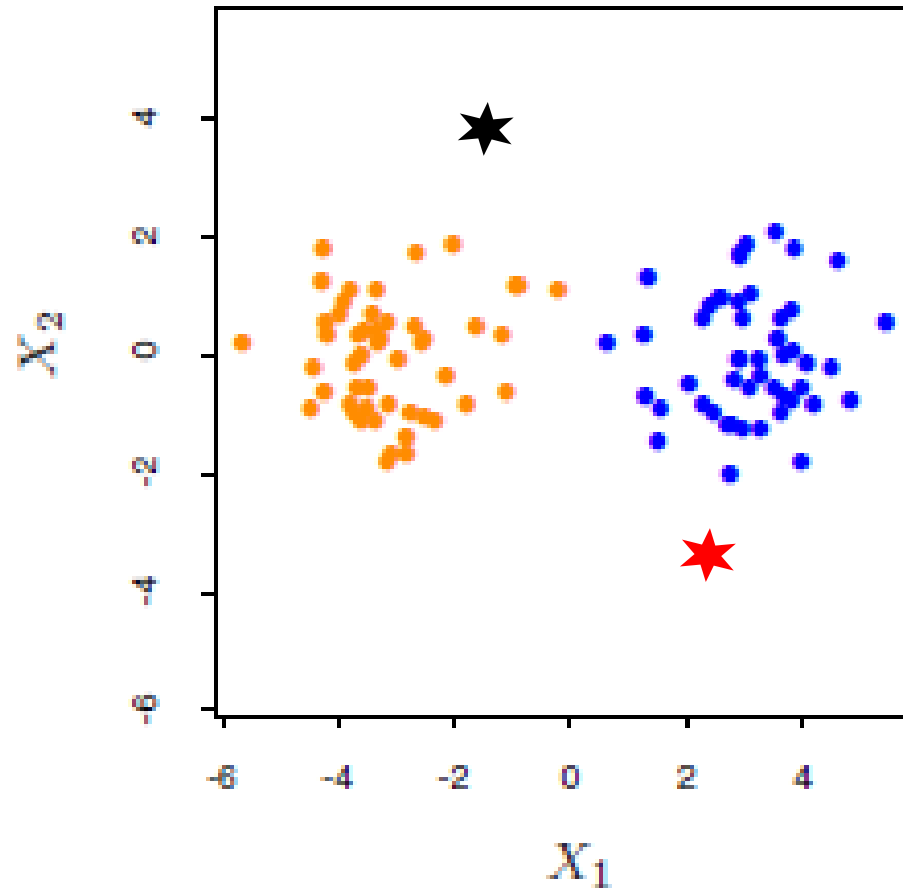
The K-means approach



$$d(x_i, x_{i'}) = \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = ||x_i - x_{i'}||^2$$

The K-means approach

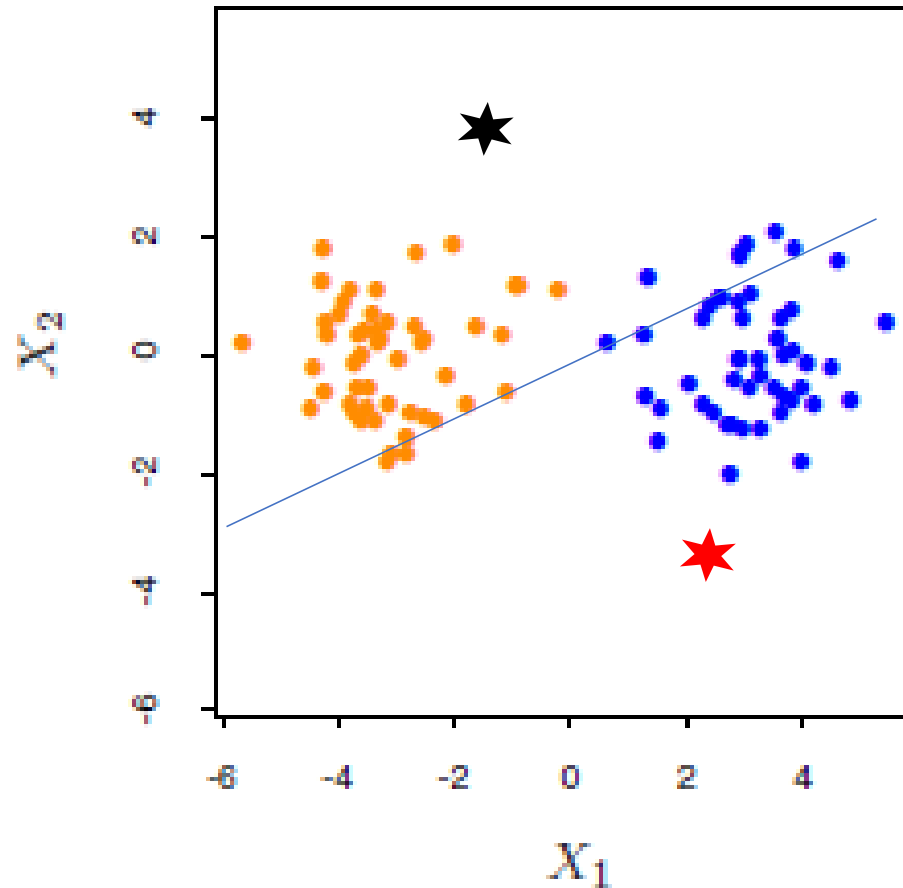
Set the coordinates at random



$$d(x_i, x_{i'}) = \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = \|x_i - x_{i'}\|^2$$

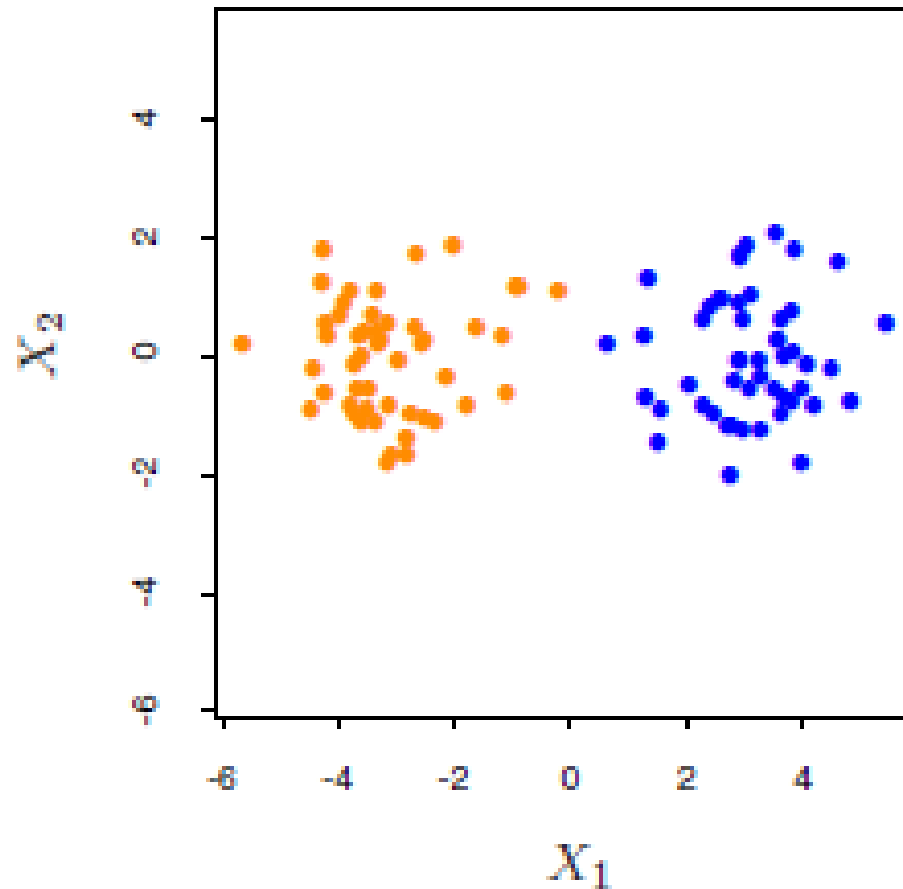
The K-means approach

Define close neighbours

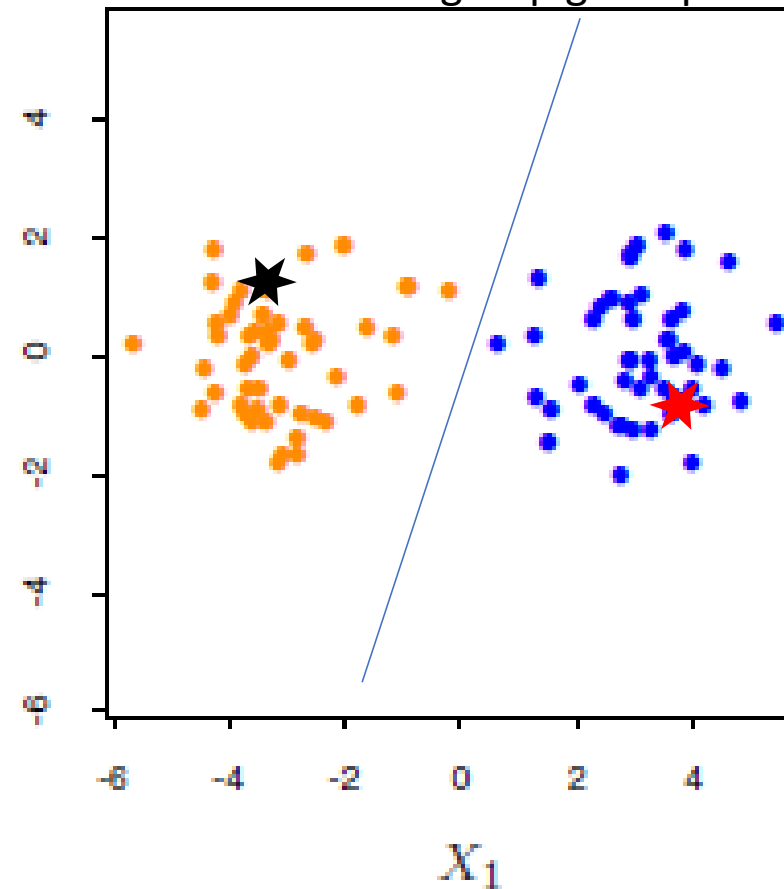


$$d(x_i, x_{i'}) = \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = \|x_i - x_{i'}\|^2$$

The K-means approach

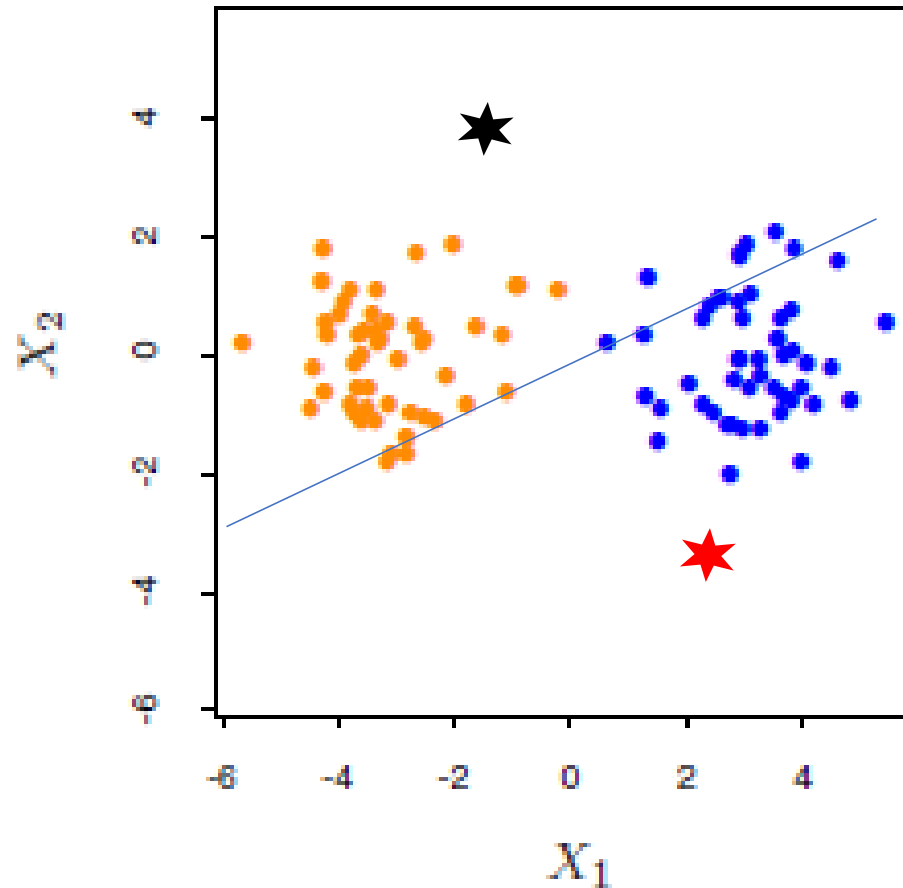


Compute the new coordinates as the mean of the points
That fall in each group given previous coordinates

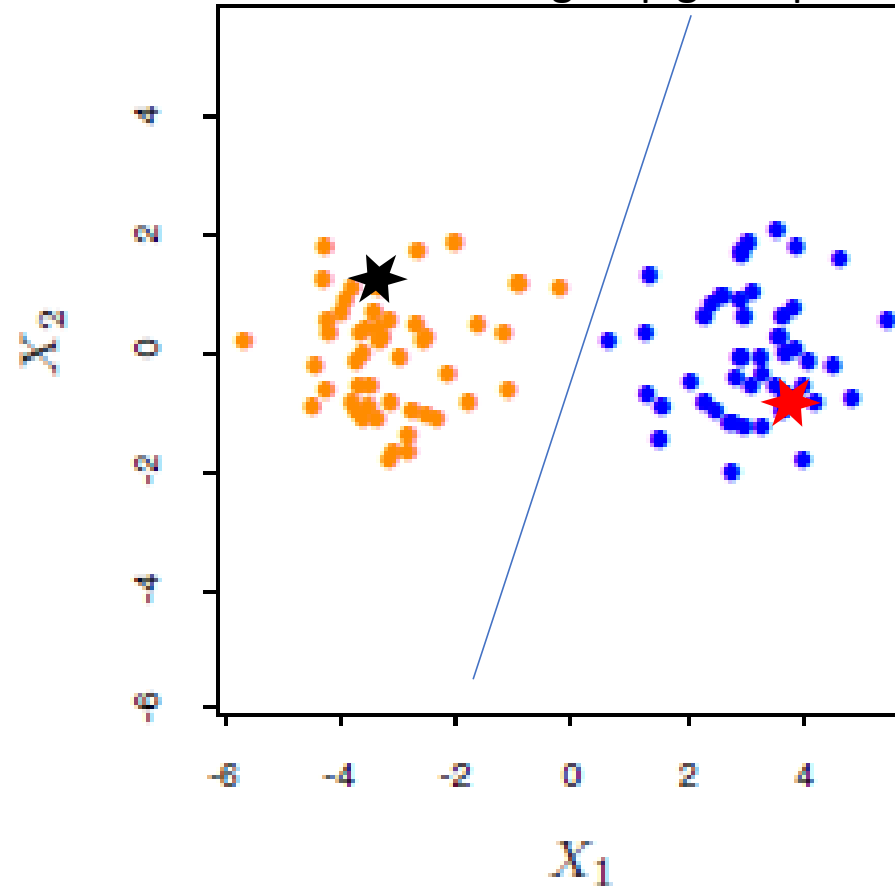


The K-means approach

Set the coordinates at random



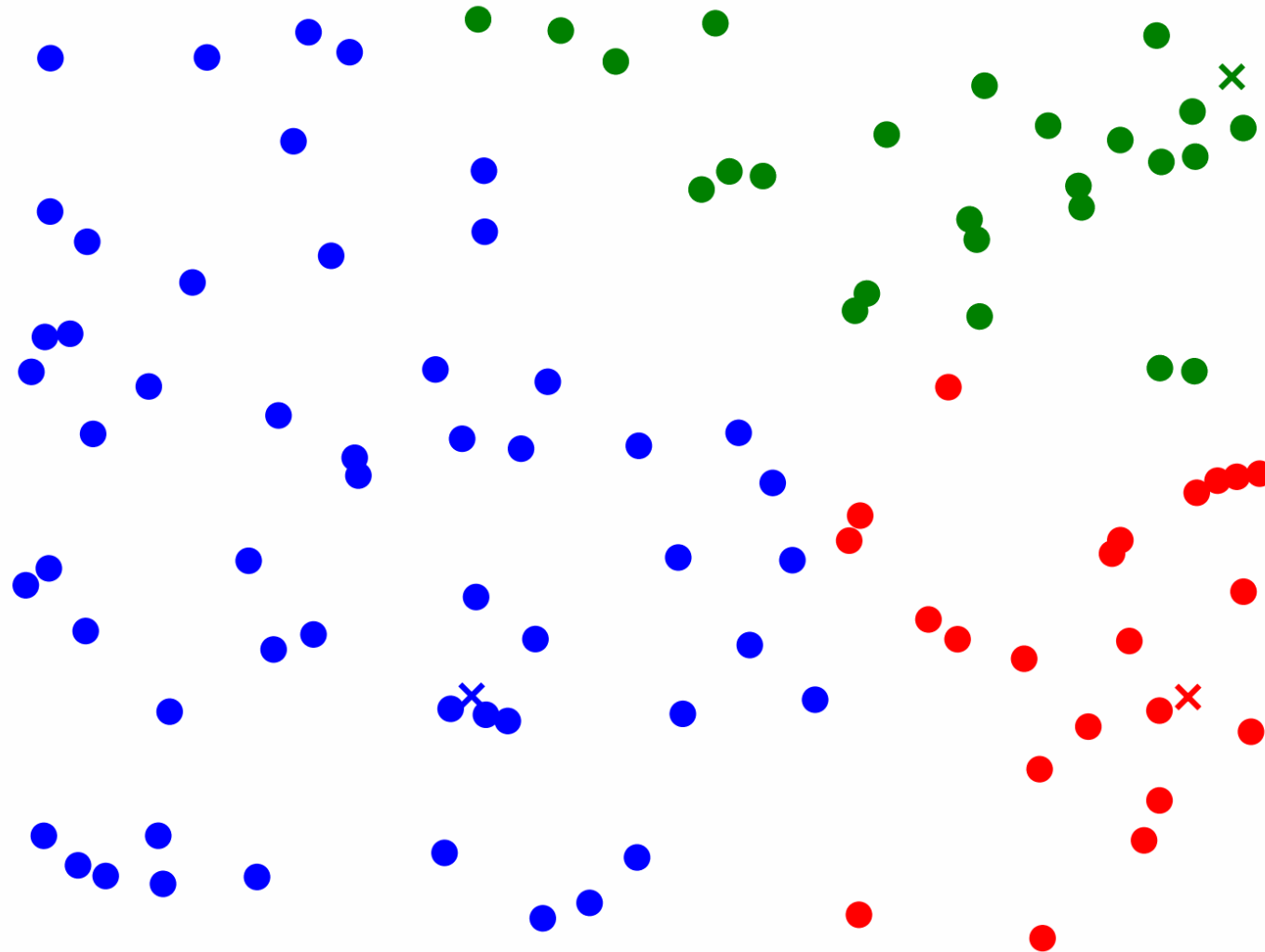
Compute the new coordinates as the mean of the points
That fall in each group given previous coordinates



$$d(x_i, x_{i'}) = \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = ||x_i - x_{i'}||^2$$

K-means.

Iteration 1: Assignment



K-means.

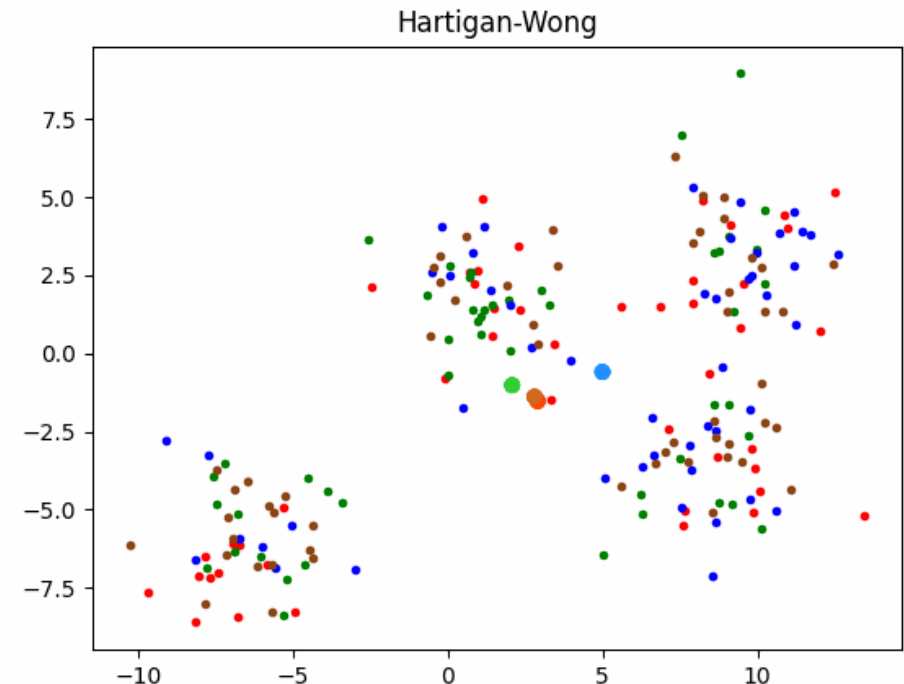
- K-means is a greedy algorithm.
- Clusters depend not only on the nature of the data, but also on which **hyperparameters** you use.

K-means.

- Initialization points
 - **Hartigan and Wong:** points which are well separated and have a large number of points within their surrounding multi-dimensional sphere can be good candidates for initial points.

Algorithm 1 Hartigan-Wong

```
1: choose  $k$  as the number of centroids
2: randomly assign all points to a centroid
3: calculate the centroids as mean of their assigned points
4: repeat
5:   for each datapoint  $d$  do
6:     for each centroid  $c$  do
7:       assign datapoint  $d$  to centroid  $c$ 
8:       compute the sum of squared distances from each point to its
         centroid
9:     end for
10:    assign  $d$  to the centroid which resulted in the smallest sum
11:    recalculate centroids as mean over all points assigned
12:  end for
13: until convergence
```



K-means.

- Initialization points
 - **Milligan**: Apply first agglomerative hierarchical clustering
 - **K-Means++**: First centroid is selected at random. The next centroid selected is the one which is farthest from the currently selected centroid. This selection is decided based on a weighted probability score. The selection is continued until we have K centroids and then K-means clustering is done using these centroids.
 - ***Put yours here***

The k-means approach

When to stop

- As the number of clusters K is changed, the cluster memberships can change in arbitrary ways (things that before were not in the same cluster at $k-1$, can be in the same cluster at k !)
- Which is the optimal number of clusters????

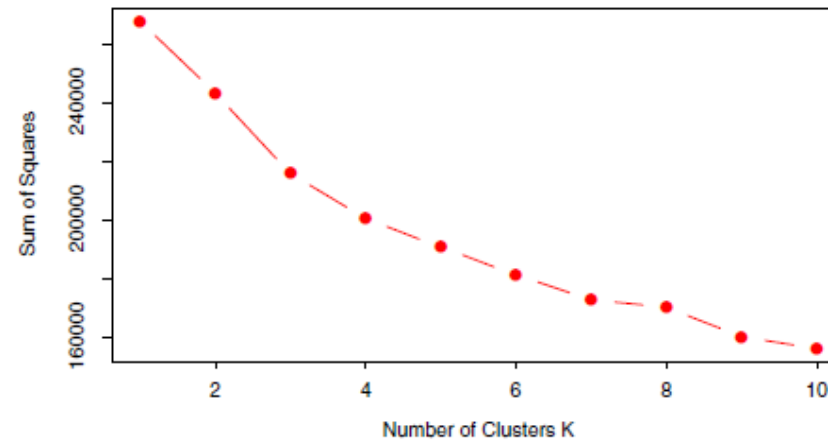


FIGURE 14.8. Total within-cluster sum of squares for K -means clustering applied to the human tumor microarray data.

The k-means approach

- **Calinski–Harabasz Index:**

The diagram shows the Calinski-Harabasz Index formula with several arrows pointing to its components for explanation:

$$CH(K) = \frac{\frac{B(K)}{(K-1)}}{\frac{W(K)}{N-K}}$$

- An arrow points from the text "Do we want to maximize or minimize this function?" to the $CH(K)$ term.
- An arrow points from the text "Distance of points Between clusters" to the $B(K)$ term in the numerator.
- An arrow points from the text "Distance of points within clusters" to the $W(K)$ term in the denominator.
- An arrow points from the text "Number of points" to the N term in the denominator.
- An arrow points from the text "Number of clusters" to the K term in the denominator.

The k-means approach

- A silhouette approach

Let us first define the numbers $s(i)$ in the case of dissimilarities. Take any object i in the data set, and denote by A the cluster to which it has been assigned. (For a concrete illustration, see Fig. 1). When cluster A contains other objects apart from i , then we can compute

$a(i)$ = average dissimilarity of i to all other objects of A .

In Fig. 1, this is the average length of all lines within A . Let us now consider any cluster C which is different from A , and compute

$d(i, C)$ = average dissimilarity of i to all objects of C .

In Fig. 1, this is the average length of all lines going from i to C . After computing $d(i, C)$ for all clusters $C \neq A$, we select the smallest of those numbers and denote it by

$b(i) = \text{minimum}_{C \neq A} d(i, C)$.

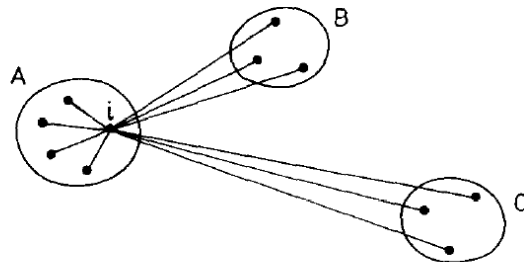


Fig. 1. An illustration of the elements involved in the computation of $s(i)$, where the object i belongs to cluster A .

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

$$-1 \leq s(i) \leq 1$$

The k-means approach

- Gap Statistic: Generate B different datasets. Compute within cluster sum of squares (W_b^*).

$$Gap(K) = \frac{1}{B} \times \sum_b \log(W_b^*(K)) - \log(W(K))$$

Pick the number of clusters K so:

$$Gap(K) \geq Gap(K+1) - s_{k+1}$$

where s_{k+1} represents the estimate of standard deviation of $\log(W_b^*(K+1))$.

- Many others!

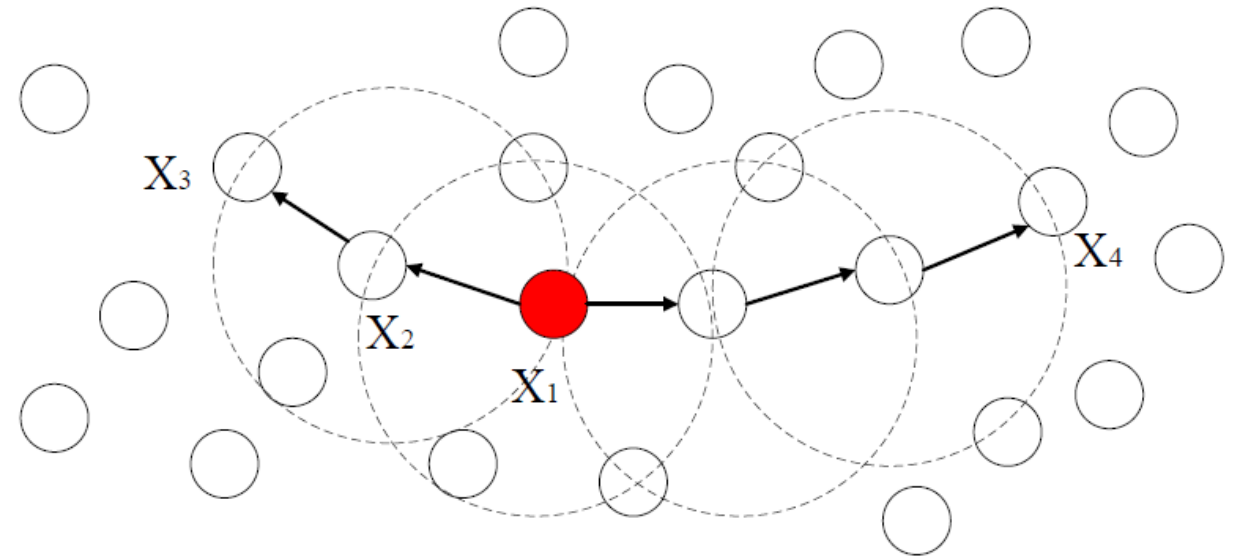
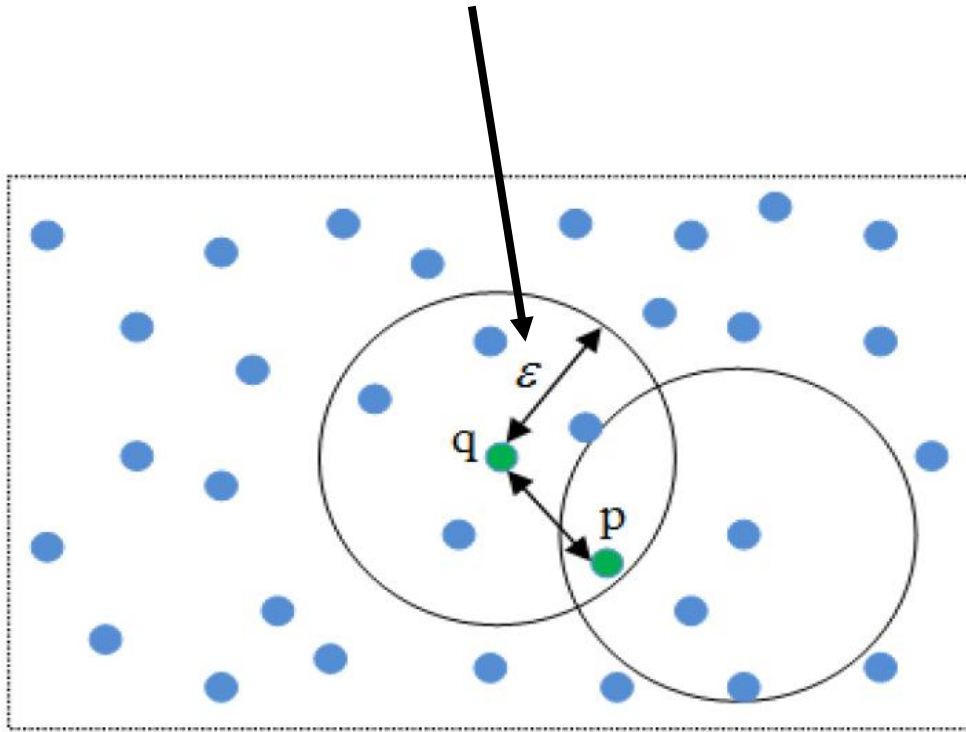
Density based approaches

- Points from the same cluster show the same density behaviour
 - Density is model free and must be identified by the algorithm
 - A Density-Based Spatial Clustering (DBSCAN)
 - Density model is pre-defined by the user
 - Mixture of probability distributions

DBSCAN

Two points belong to the same cluster if there is a density reachable path connecting them

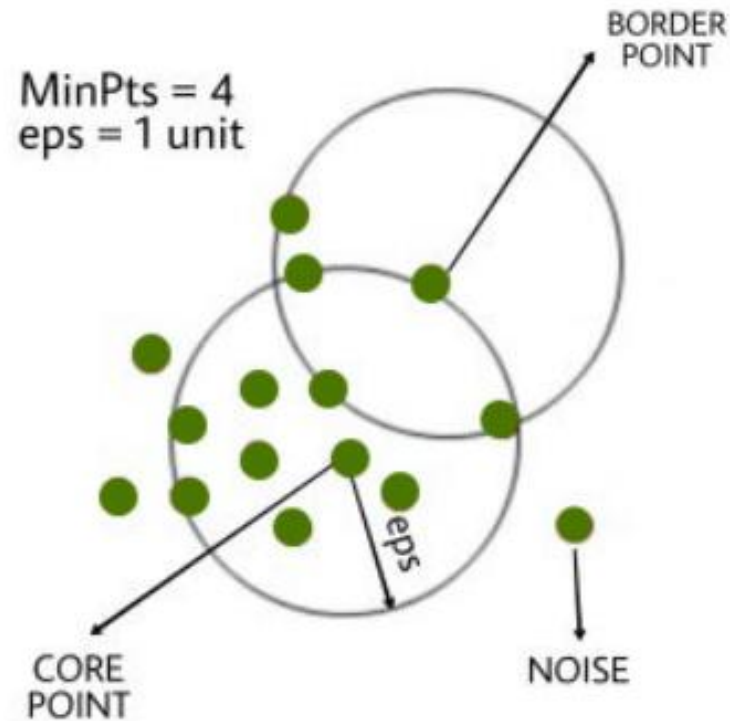
Hyperparameter. Eps is the radius of the adjacent neighborhood of a considered data point



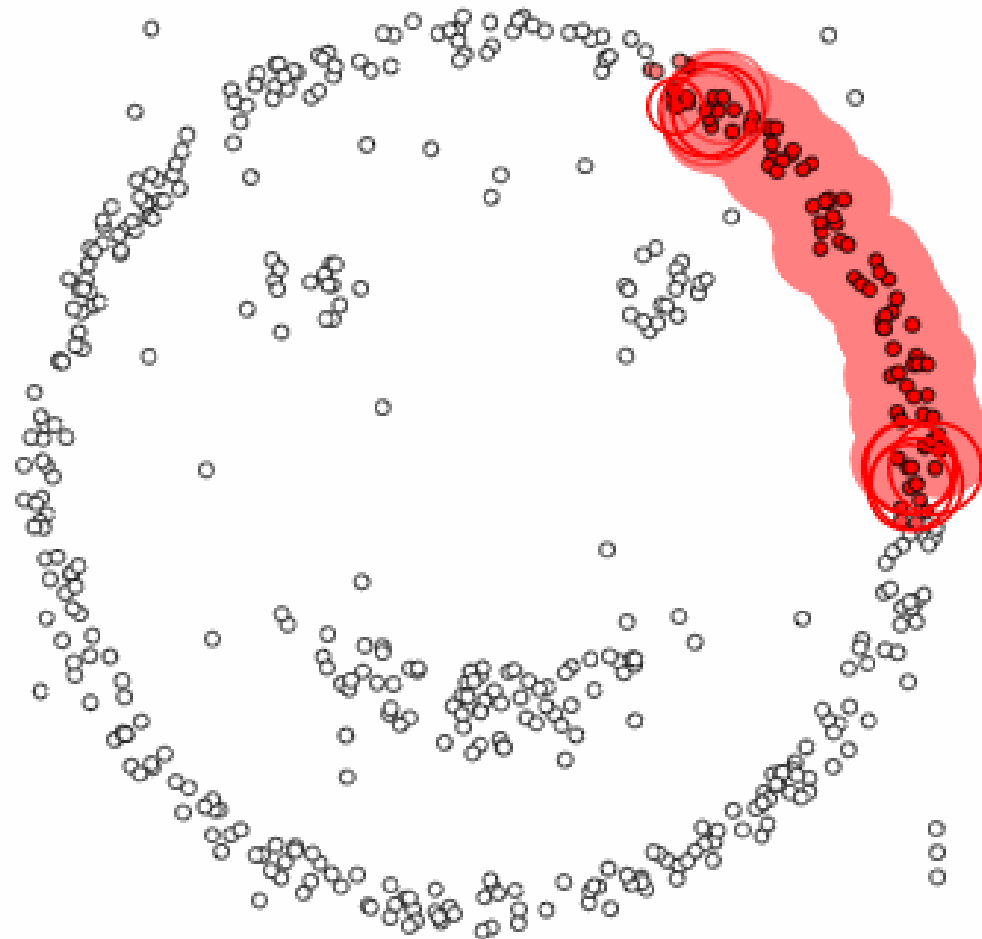
DBSCAN

Two points belong to the same cluster if there is a density reachable path connecting them

Hyperparameter. MinPts is the adjacent minimum number of data points located in the given region.

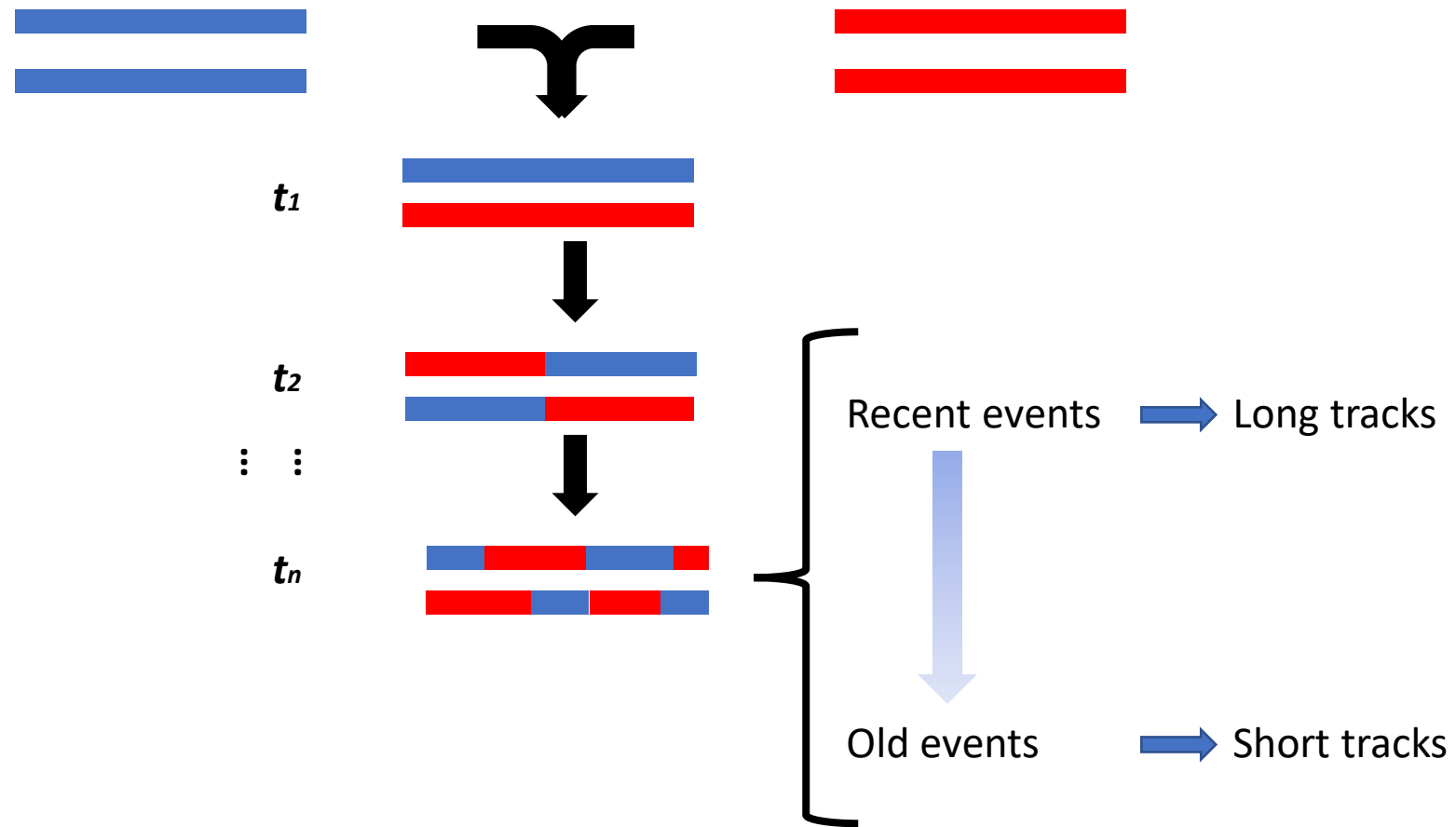


DBSCAN

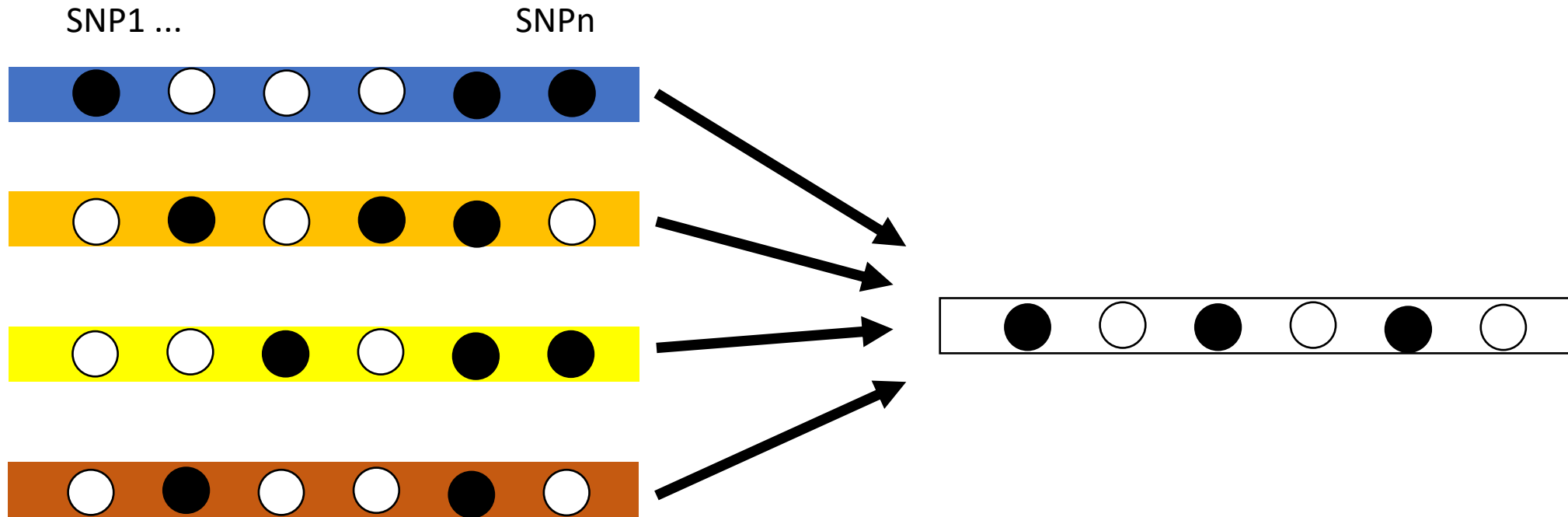


epsilon = 1.00
minPoints = 4

Change how data are defined. Local Ancestry

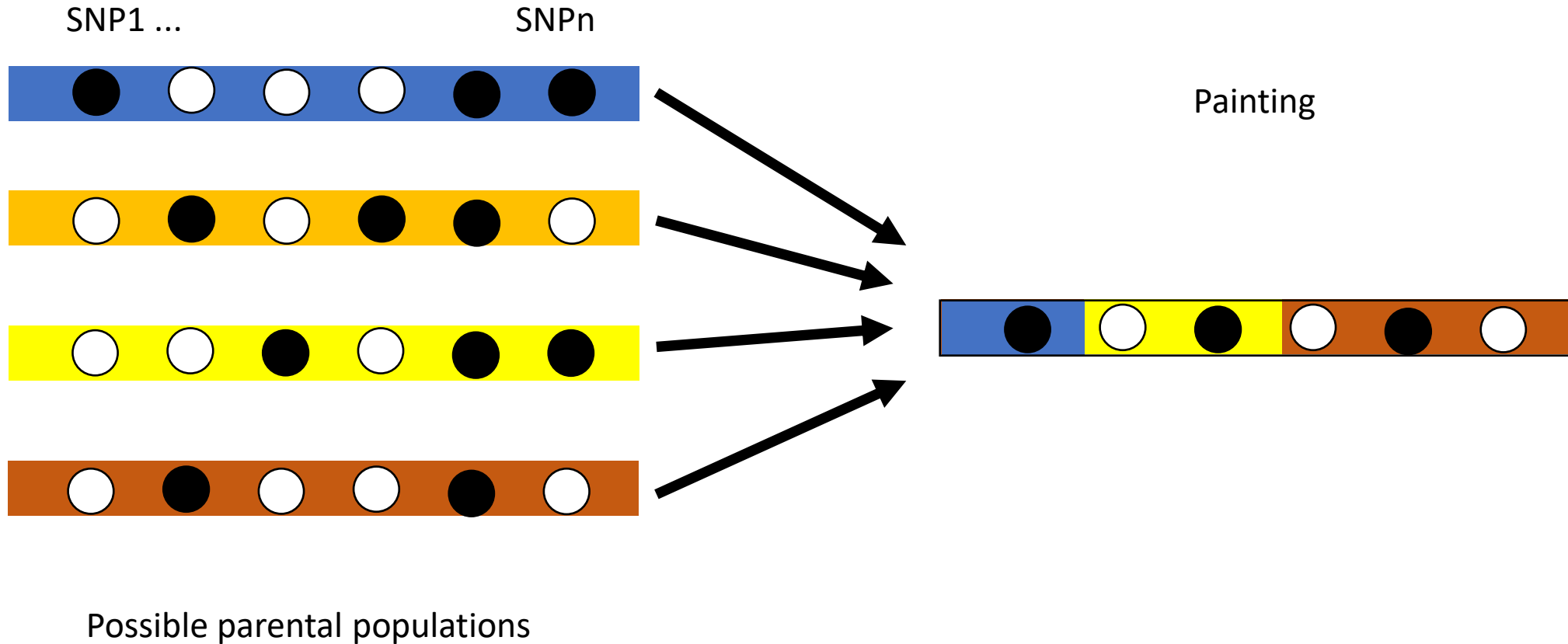


Change how data are defined. Local Ancestry



Possible parental populations

Change how data are defined. Local Ancestry



Local Ancestry. Estimated with Machine Learning methods

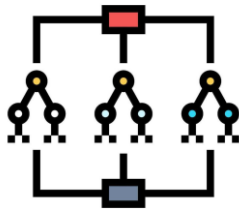
window

RFMix: Random Forest + Hidden Markov Model

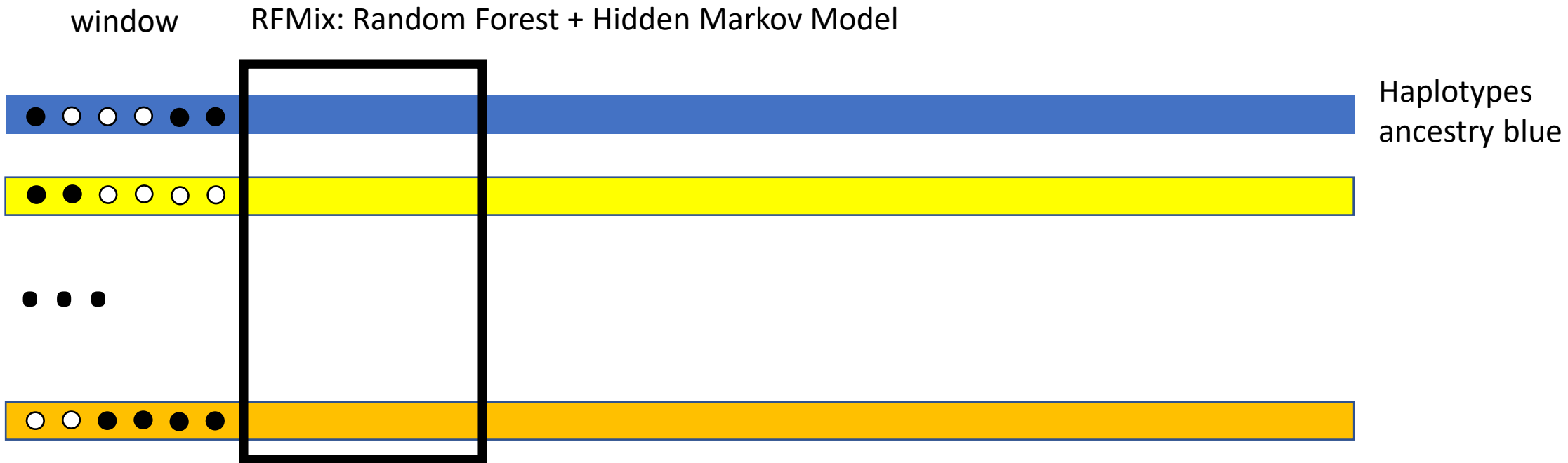


Haplotypes
ancestry blue

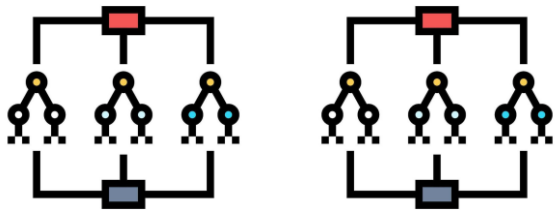
RF model
combining
multiple binary
decision trees
for predicting
ancestry



Local Ancestry. Estimated with Machine Learning methods



RF model
combining
multiple binary
trees for
predicting
ancestry



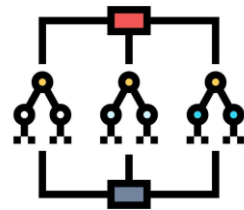
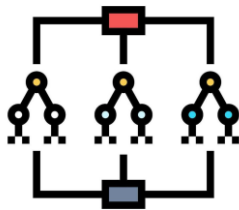
Local Ancestry. Estimated with Machine Learning methods

window

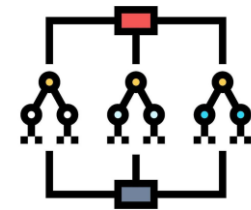
RFMix: Random Forest + Hidden Markov Model



RF model
combining
multiple binary
trees for
predicting
ancestry



...



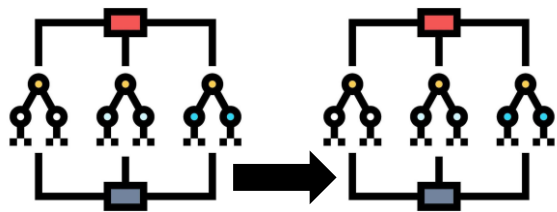
Local Ancestry. Estimated with Machine Learning methods

window

RFMix: Random Forest + Hidden Markov Model

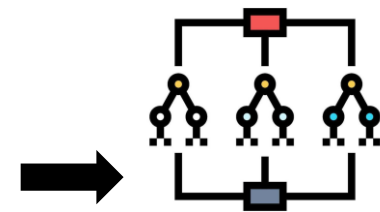


RF model
combining
multiple binary
trees for
predicting
ancestry

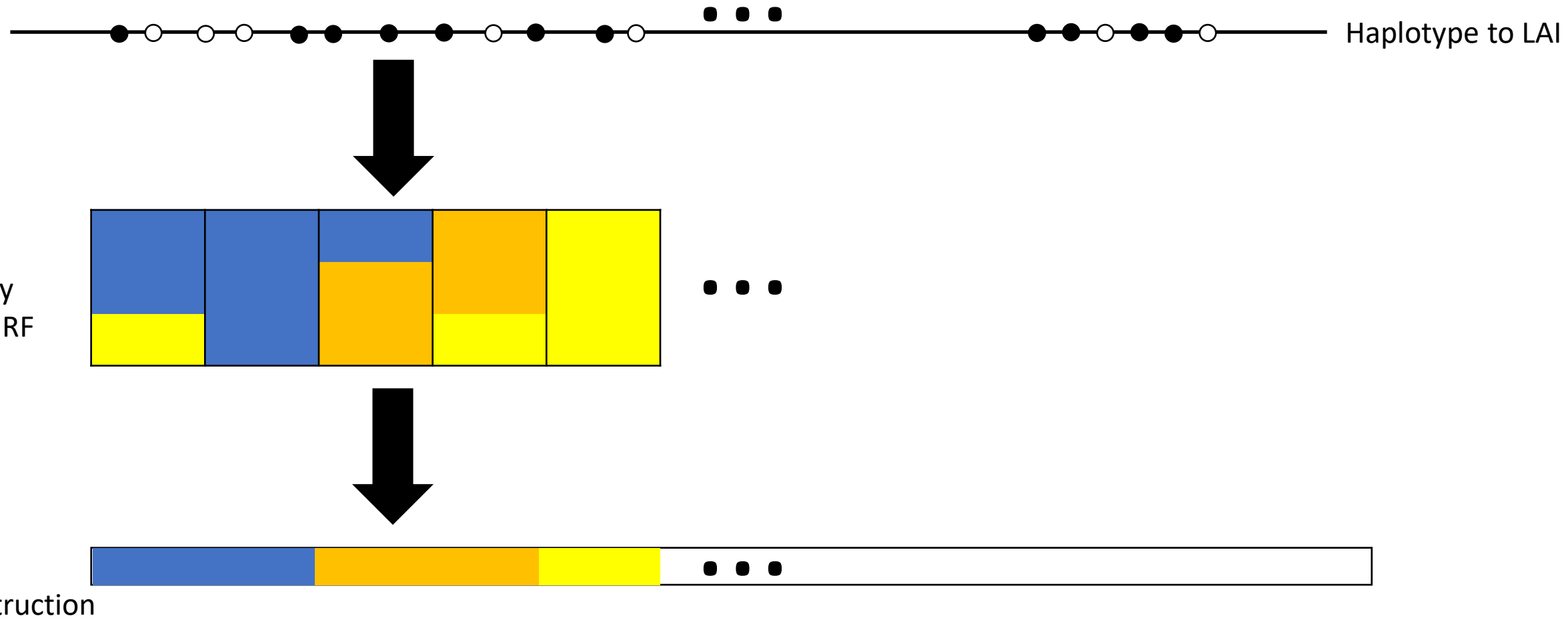


HMM

...

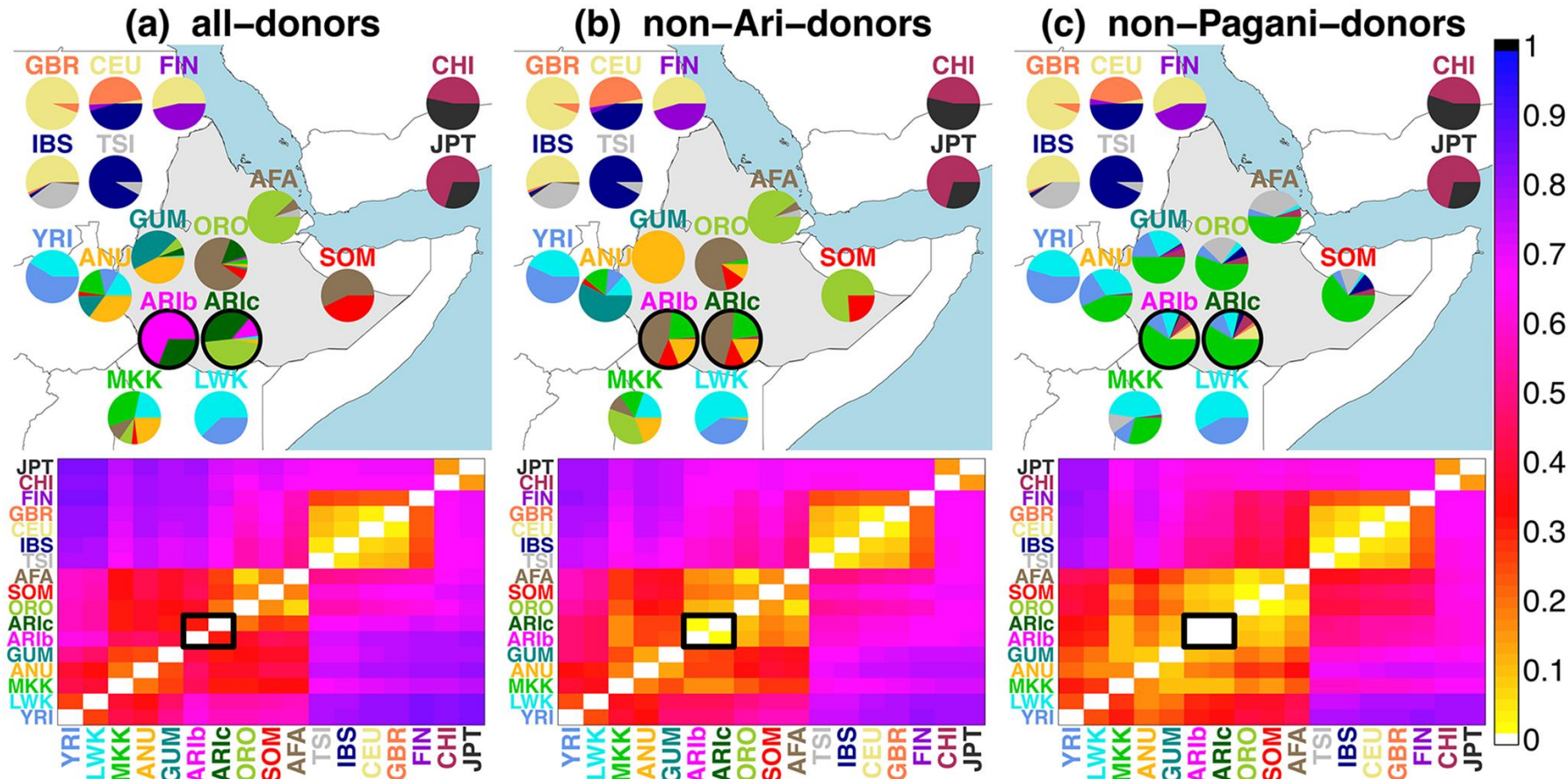


Local Ancestry. Estimated with Machine Learning methods



Change how data are defined. Local Ancestry

Chromopainter. Define who are the parents. Non-supervised: all; supervised: some-donnors

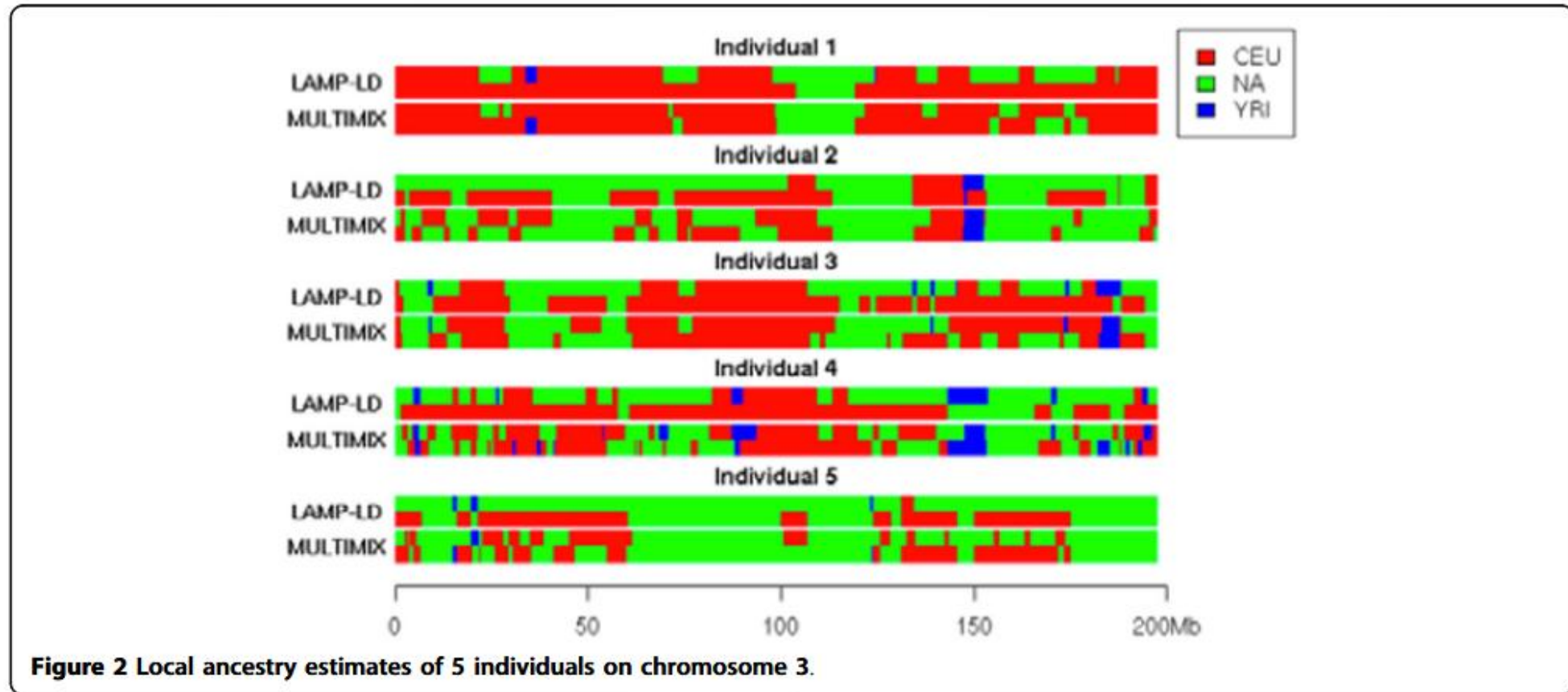


Local ancestry algorithms

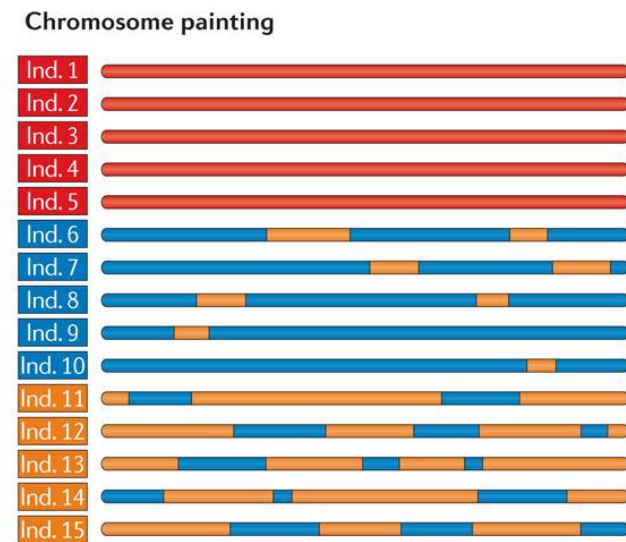
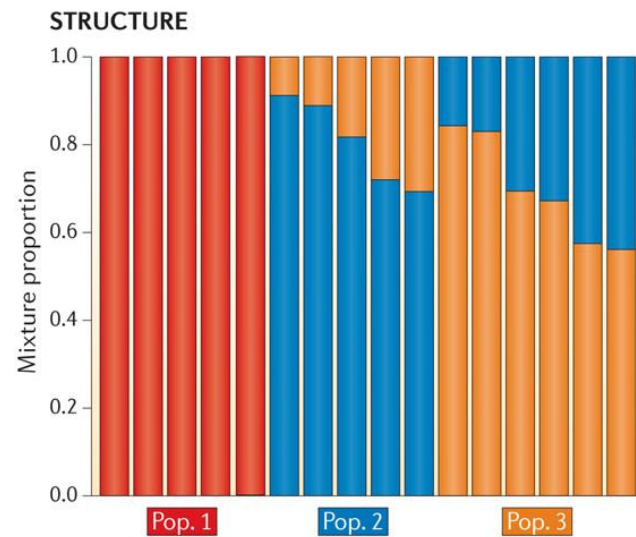
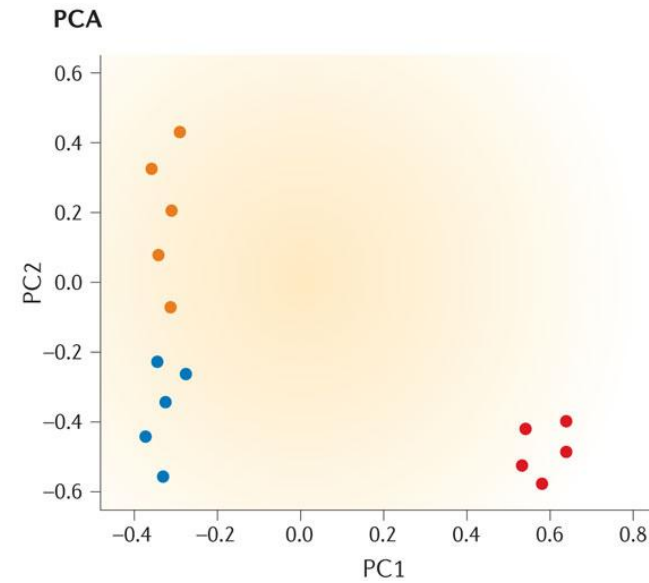
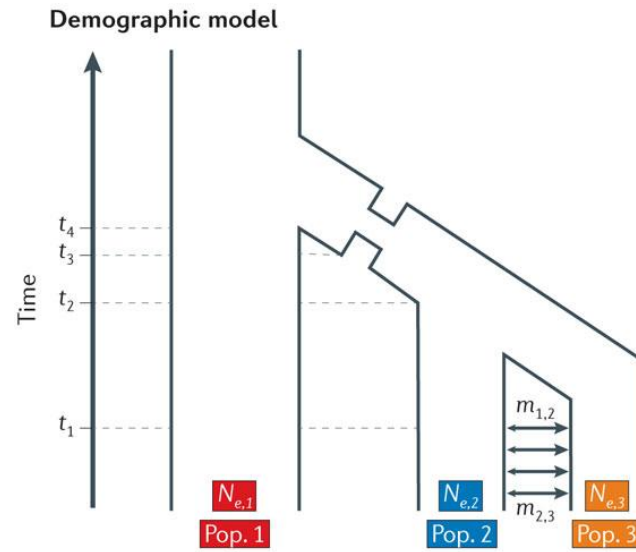
Tool / Algorithm	Key Features	Input	Output	R Package / Software
RFMix	HMM-based, phased data	Phased VCF + reference panel	Local ancestry per SNP	RFMix v2
ELAI	EM-based, handles multiple admixture pulses	Unphased genotypes	Ancestry probabilities	ELAI GitHub
LAMP / LAMP-LD	Window-based method; works with admixed populations	Genotype matrix	Ancestry blocks	LAMP (legacy)
Ancestry_HMM	HMM using recombination info, includes admixture timing	Phased haplotypes	Local ancestry + admixture time	Ancestry_HMM GitHub
HAPMIX <i>(deprecated)</i>	Two-way admixture model using HMM	Phased genotypes	Local ancestry states	HAPMIX (legacy)
WINPOP	PCA in sliding windows for non-model organisms	Unphased genotypes	Window-based ancestry assignments	winPop GitHub

Local ancestry algorithms

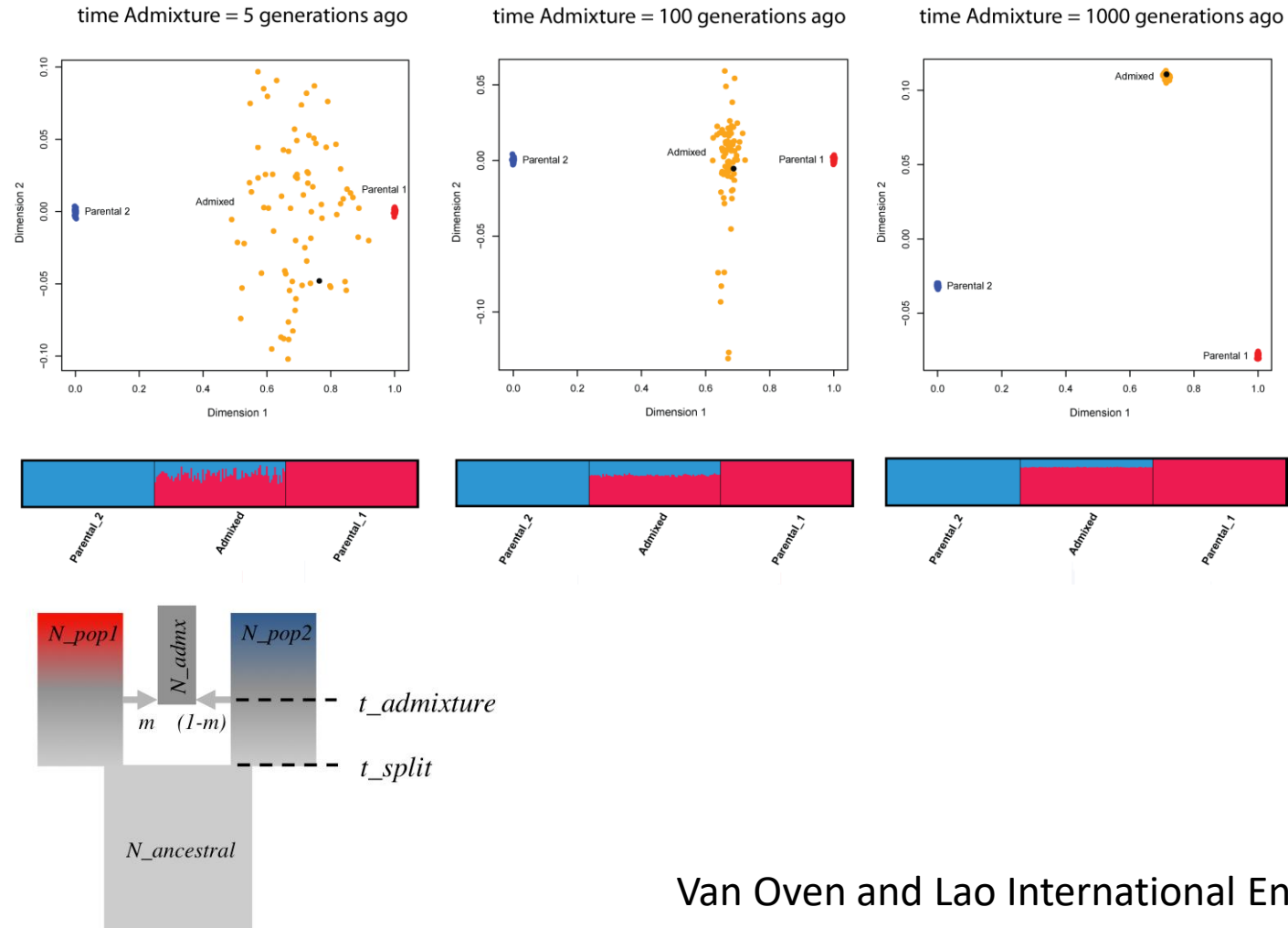
Again, estimations can differ depending on the algorithm that is applied



In summary

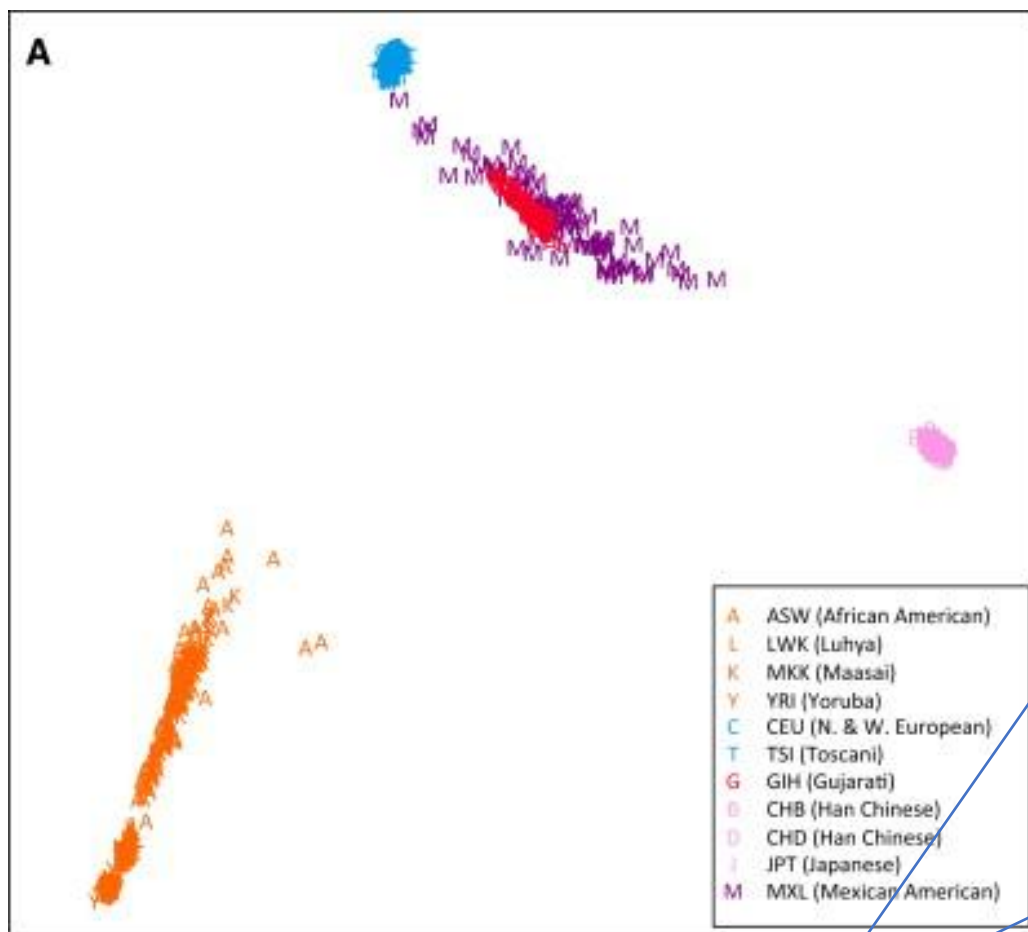


Ancestry estimation and demographic parameters?

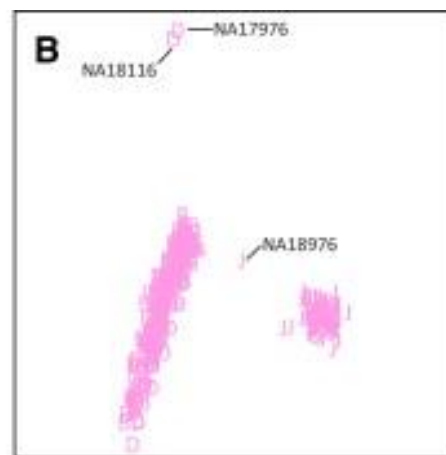


Ancestry estimation and genetic relationship?

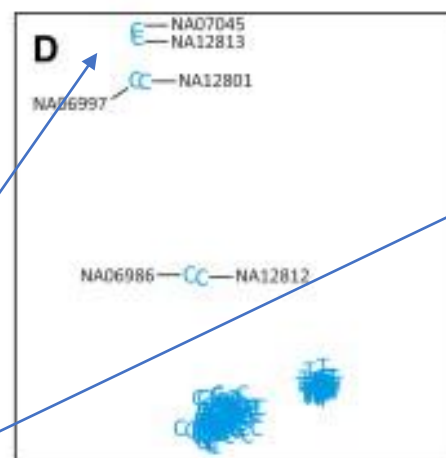
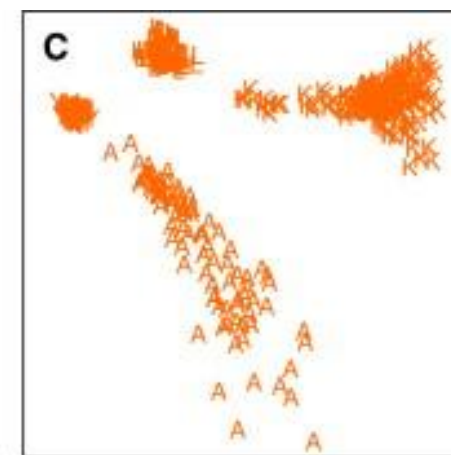
MDS



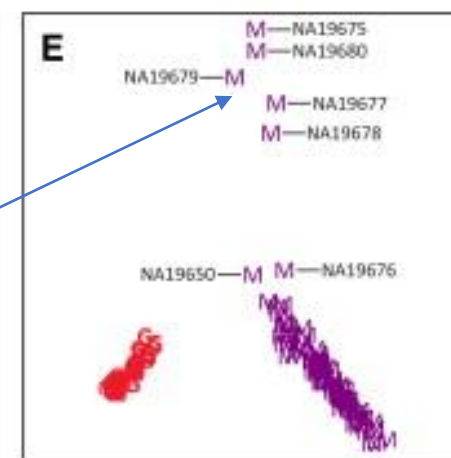
East Asian descent
(CHB, CHD, and JPT)



African descent
(ASW, LWK, MKK, and YRI)



European descent
(CEU and TSI).

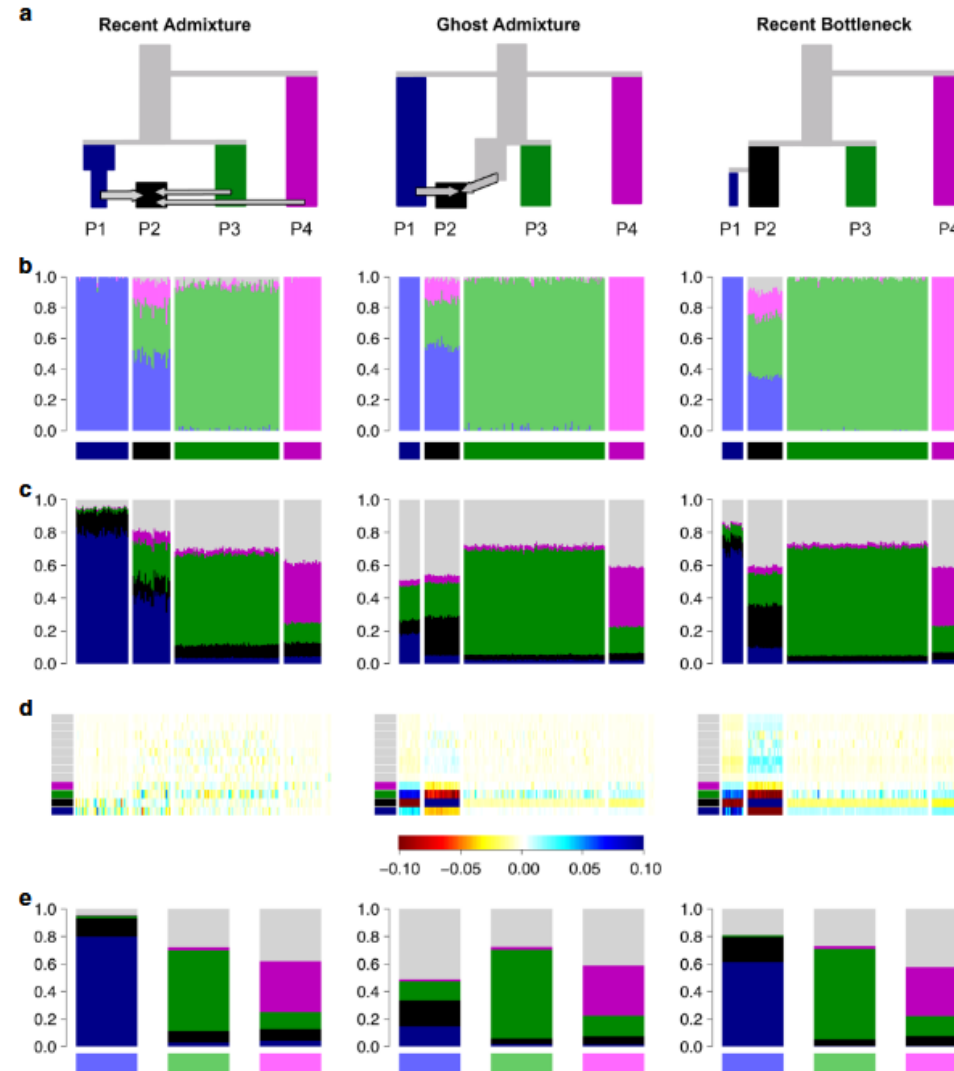


MXL and GIH.

related individuals

Ancestry estimation and demographic parameters?

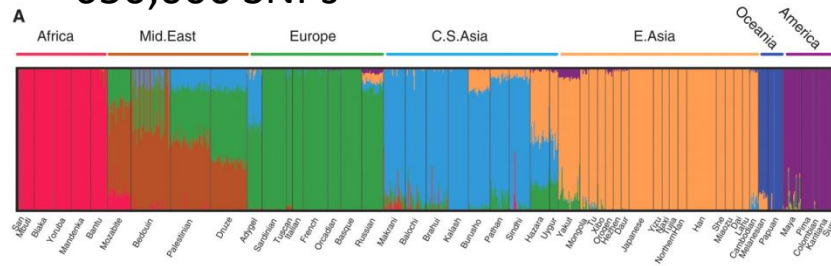
Lawson, van Dorp & Falush. Nat Comms. (2018).



Interpretation

FRAPPE

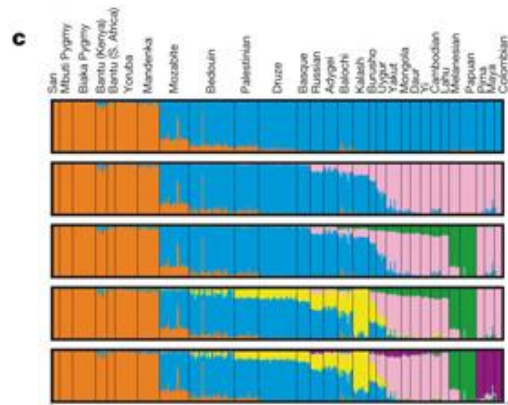
650,000 SNPs



Li et al Science 2008

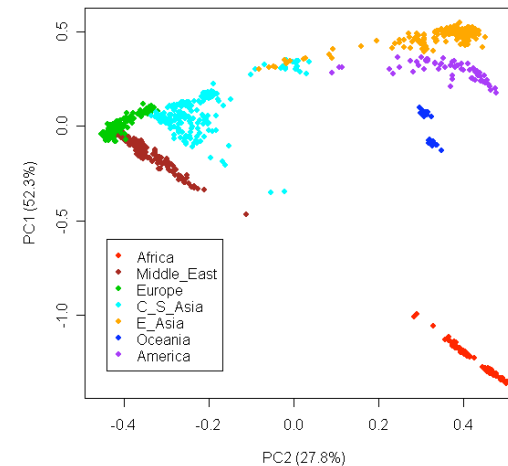
550,000 SNPs

STRUCTURE



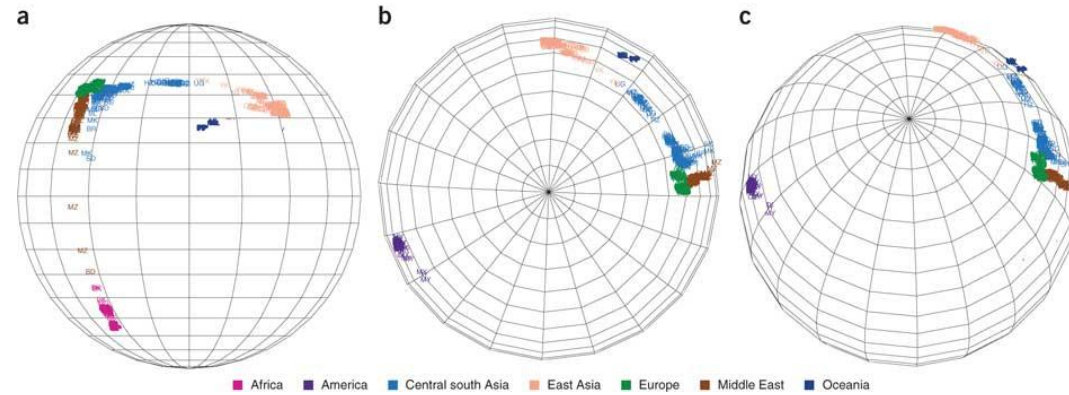
Jakobsson et al Nature 2008

HGDP-CEPH panel

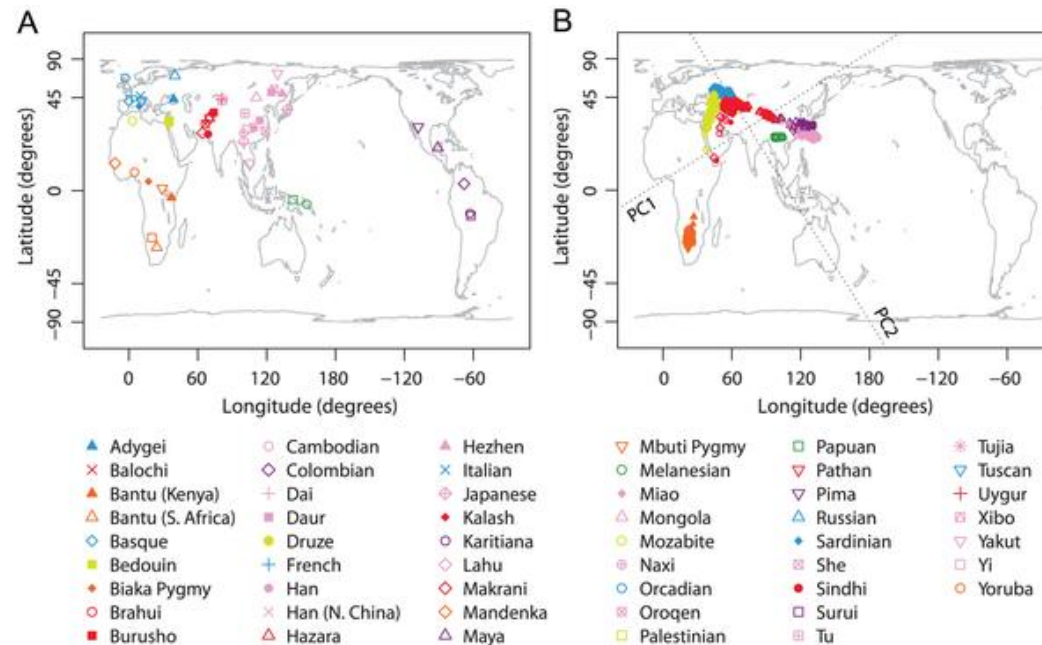


Li et al Science 2008

Interpretation

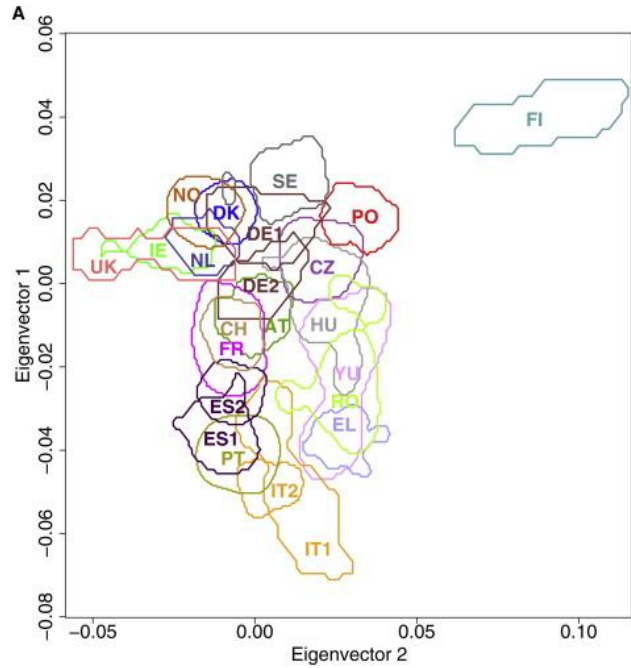


Yang et al (2012) Nature Genetics

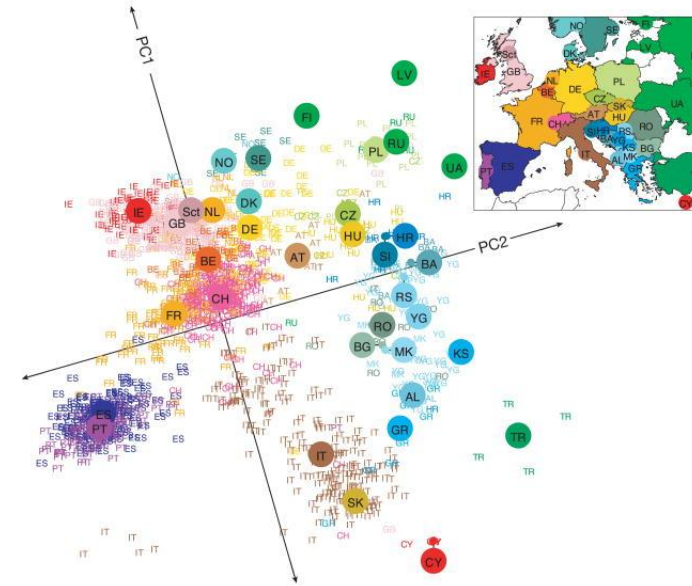
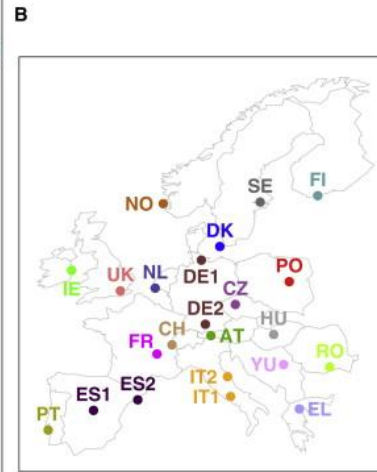


Wang C et al (2012) PLoS Genet

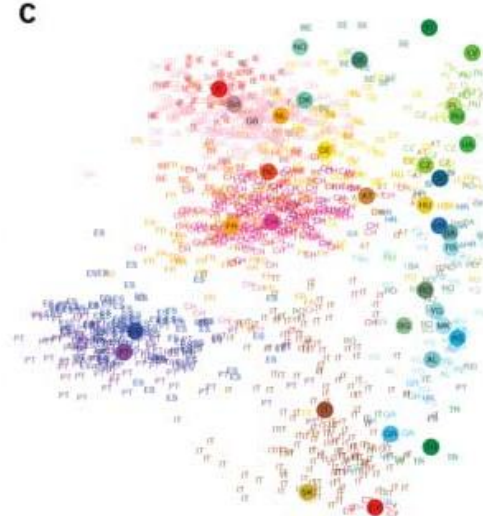
European genetic substructure



Lao et al 2007 Current Biology



Novembre et al 2007 Nature



Yang et al (2012) Nature Genetics

Takehome messages

- Ancestry estimation is a subfield of Machine learning
- Feature extraction: Unsupervised learning methods that compresses information of features. In the case of genetic data, genetic variation. PCA, ADMIXTURE, sNMF...
- Demographic interpretation can be challenging. BEWARE STORY TELLING!
- Local ancestry focus on LD. They are supervised or semi-supervised machine learning methods. They usually build ML models at loci and HMM to connect adjacent loci
- Local ancestry can be used to identify recent admixture events
- REMEMBER: results will depend on the assumptions of the algorithms!

Best-practice workflow

- Define the biological question
- Decide the model (spatial?, ancestral proportions?, etc)
 - Decide the hyperparameters
- Data preprocessing:
 - Remove/impute markers with missing data
 - Remove markers out of HWE
 - Remove markers at low frequency
 - Remove strongly related individuals
- Remove regions with large LD (inversions, etc)
- Control by LD (ideally within population, but since you are trying to define what is a population...)
- Interpret conservatively
 - SNV discovery in microarray data can also bias the relationship
 - Many possible evolutionary models can produce the same result

Thank you very much!

